Risk Sharing and Information in Migration Networks

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Abstract

This paper develops a model of rural–urban migration and social network formation under uncertainty and empirically tests its predictions using household panel data from rural Kenya. In the model, linked households pool their income and reveal their migration outcomes to one another. Households choose links endogenously by comparing the value of new information about the city to the net gains (or costs) from informal network-based insurance and redistribution. The model predicts that households may avoid forming potentially informative links when expected redistribution costs are high, slowing the learning process. Using detailed household data on migration beliefs, decisions, and social connections, we find empirical support for several of the model's key comparative statics.

JEL Classifications: O15, D85, D83, J61, D81
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1 Introduction

In many developing countries, migration from rural to urban areas offers households a path toward higher income, but is often risky and costly. When migration returns are uncertain and experimenting is difficult, households often rely on social networks to form beliefs about their potential migration outcomes. These networks, however, serve dual and sometimes conflicting roles: they facilitate information transmission through past migrants, while also imposing obligations to share income (Jakiela and Ozier, 2016, Baseler, 2023). The result is a tension in how households decide to develop new social relationships: households may want to link with others to improve their information about migration outcomes, but doing so exposes them to informal taxation via redistribution.

We study the joint decisions made by rural households about migration and social network links. To do so, we formulate and solve a model that marries existing frameworks from two literatures: one studying migration decisions under uncertainty and another examining static risk sharing within social networks. In our model, risk-averse rural households choose whether to migrate in each of two periods and may form new bilateral network links. Urban income contains a persistent common component that households do not observe directly, implying that one household's migration outcome provides information about the returns available to others. Households learn about others' migration outcomes only when they are socially linked, and forming new links is costly. At the same time, linked households share both rural and urban income, so an additional connection provides insurance, but may also require a household to transfer income to their poorer neighbors. Households therefore form links only when the combined informational and risk-sharing benefits exceed the cost of link formation, net of the expected cost or benefit of redistribution.

Solving the model reveals how the informational and financial consequences of social links interact with migration incentives, generating a set of comparative statics that we take to the data. Households are less likely to form links when those links require greater expected redistribution obligations. Greater uncertainty about migration returns increases the value of a information seeking link when the household would otherwise migrate, while the effect is ambiguous when it would not migrate. Migration experience makes households valuable sources of information, increasing their attractiveness as providers of migration-related advice. Information obtained from outside social networks substitutes for network-based learning, and so reduces the value of new links and changes migration decisions the most for those with few social connections. Finally, migration becomes more likely when expected urban returns are greater or the rural outside option worse, including when rural income streams are riskier.

We then test these predictions on data collected from the large-scale panel study of Barnett-Howell et al. (2026). The study followed approximately 16,800 households across 560 villages in rural Kenya over four survey waves, collecting repeated measures of migration beliefs and intentions, realized migration and income outcomes, and households' financial and information-seeking relationships within their villages. After the baseline survey, randomly selected villages received interventions providing information about urban wages. The combination of panel network data and experimental variation allows us to examine how households' demand for social information responds to uncertainty, redistribution obligations, and externally provided information, and how these forces shape subsequent migration behavior. The surveys distinguish between financial and informational links and different directions of information exchange, allowing us to study a rich network of social relationships. We find support for several of the model's predictions, suggesting that information value and informal insurance act as competing incentives to form social links.

Related Literature. A large literature models migration as a response to spatial differences in income and costs of moving (Sjaastad, 1962, Harris and Todaro, 1970, Borjas, 1987, Kennan and Walker, 2011). In many developing countries, large rural–urban productivity and income gaps imply substantial returns to migration, while barriers to mobility contribute to the persistence of these gaps and to aggregate misallocation (Morten, 2019, Lagakos, 2020). Increasingly, research emphasizes that migration decisions depend not only on expected returns, but also on uncertainty about those returns and on the information available to potential migrants. Given that migration is costly and its returns are difficult to observe without experimenting, households may remain in rural areas even when migration is profitable (Bryan et al., 2014). Misperceptions about destination earnings and risks can consequently have important effects on migration choices (Shrestha, 2020, Baseler, 2023, Barnett-Howell et al., 2026). Social networks are a natural source of such information: previous migrants can transmit knowledge about employment opportunities, earnings, costs, and conditions at the destination, while also facilitating access to jobs and other resources (Munshi, 2003).[1] However, the informational and insurance functions of networks may interact strategically. Migrants often transmit knowledge about urban opportunities to family members and others at the origin, but may conceal favorable outcomes when reporting high earnings would generate pressure to remit or share more income (Baseler, 2023, Ambler,

[1]Which households provide this information may itself depend on network structure. Work on information diffusion shows that communities possess useful knowledge about who is especially central for spreading information and that targeting these "gossips" substantially increases diffusion (Banerjee et al., 2019). Our paper instead studies how households endogenously seek links to people who possess relevant migration experience and how the redistributive consequences of those links affect this search.

2015).[2] A closely related paper to ours is Barsbai et al. (2024), which develops a model in which externally provided information and social connections are substitutes, and shows that providing new immigrants with pre-departure settlement information reduces the number of social connections they subsequently acquire. To the best of our knowledge, our paper is the first to jointly model migration and endogenous network formation when the same social links perform two roles: they provide information about uncertain migration returns and they create informal-insurance and redistribution incentives. A more detailed comparison of our model with closely related models can be found in Appendix B.

A second literature studies informal risk sharing in rural economies, which can partly substitute for formal credit and insurance in helping households smooth consumption and cope with idiosyncratic shocks (Townsend, 1994, Ligon et al., 2002, Fafchamps, 2011). Risk sharing is generally incomplete, however, because transfer arrangements are constrained by limited commitment, private information, imperfect monitoring, and barriers to enforcing informal agreements (Ligon et al., 2002, Genicot and Ray, 2003, Kinnan, 2022).[3] The identity and position of a household's contacts therefore affect both the insurance it can obtain and the credibility and obligations of its informal arrangements (Fafchamps and Gubert, 2007, Ambrus et al., 2014, 2022). A large theoretical literature has examined how these informal arrangements arise endogenously, showing how participation depends on self-enforcement constraints, the distribution of risk, the benefits of indirect connections, and the transfers expected within the resulting network (Genicot and Ray, 2003, Bramoullé and Kranton, 2007, Bloch et al., 2008, Ambrus and Elliott, 2021). Our contribution to this literature is to theoretically and empirically study an additional incentive influencing the formation of risk-sharing networks: the value of information about urban income opportunities.

A related literature studies the relationship between migration and informal insurance. Migration can help households diversify income across locations whose shocks are imperfectly correlated, while remittances allow migrants and origin households to share resources across space (Stark and Levhari, 1982, Rosenzweig and Stark, 1989). Remittances can reflect altruism, exchange, or reciprocal insurance, and may insure either the household remaining at the origin or the migrant against shocks at the destination (Lucas and Stark, 1985, Batista and Umblijs, 2016, Bourlès et al., 2017). Informal insurance can facilitate migration by protecting households against adverse outcomes, but the same arrangements can also discourage

[2]This mechanism is related to a broader literature showing that demands from relatives and community members can induce households to conceal resources, borrow in order to appear poor, or otherwise alter their economic choices to avoid "kinship taxation" (Baland et al., 2011, Jakiela and Ozier, 2016, Squires, 2016).

[3]Social proximity can help overcome some of these constraints, for example by facilitating contract enforcement and supporting greater risk sharing (Attanasio et al., 2012, Chandrasekhar et al., 2018).

migration when leaving weakens access to village insurance or when migrants anticipate substantial claims on their urban earnings (De Weerdt and Hirvonen, 2016, Munshi and Rosenzweig, 2016, Morten, 2019, Meghir et al., 2022). Much of this literature treats these social relationships as fixed. Our paper brings these literatures together by endogenizing social network formation as a function of migration information and income sharing.

The remainder of the paper is organized as follows. Section 2 presents the theoretical framework and characterizes households' optimal migration and link-formation decisions. Section 3 describes the empirical setting, data, and measurement of key variables, and states the key empirical analogs of our model's theoretical predictions. Section 4 presents our empirical results, and Section 5 concludes. Additional proofs and results are provided in the Appendix.

2 Theoretical Framework

We model the migration decisions and endogenous network formation of households embedded in a rural risk-sharing network. Each rural household makes a binary decision to migrate or not migrate to a single urban location based on its beliefs about the distribution of potential migration income. Households can form new network links but face a trade-off: new links can improve information about urban income and improve risk sharing, but can also create additional income-sharing obligations. We model two periods: households choose whether to migrate in each period and may form new links between the two migration decisions.

2.1 Model Environment

Utility. Households' preferences are represented by a constant absolute risk aversion (CARA) utility function,

$$u(c_{it}) = -e^{-rc_{it}},$$

where c_{it} is household i's consumption at period t and $r > 0$ is the coefficient of absolute risk aversion. Consumption is a function of rural income, urban income, and transfers, as defined below.

Rural Income. In each period $t \in \{1, 2\}$, households earn rural income y_{it} which is drawn independently from a normal distribution:[4]

$$y_{it} \sim \mathcal{N}(\bar{y}, \sigma^2).$$

Migration and Urban Income. Migration is a binary choice, $M_{it} \in \{0, 1\}$, made in each period, with an exogenously given migration cost $q > 0$. Migration choices are observable and migration lasts for one period. Households that migrate receive, in addition to their rural income, urban income

$$e_{it} = z_t + \tilde{e}_{it},$$

where z_t is a common component and $\tilde{e}_{it} \sim \mathcal{N}(0, \sigma^2_{\tilde{e}})$ is an idiosyncratic shock which is independent and identically distributed across households and time. The common component in the second period is

$$z_2 = \rho z_1 + \lambda_2,$$

where $\lambda_2 \sim \mathcal{N}(0, \sigma^2_\lambda)$ and is independent of z_1. Persistence $\rho \in (0, 1)$ in z_t makes current migration outcomes informative about future ones.

Social Network. We consider an initial, exogenously given and observable adjacency matrix G_1 associated with the network graph $\mathcal{G}_1$. The network is undirected and unweighted.[5] For each household i, let N_{i1} denote the set of its neighbors (we will later interpret this set as a rural village) and let $d_{i1} = |N_{i1}|$ denote its degree. We define $\bar{N}_{i1}$ as the set of neighbors including the household itself.

Beliefs and Information. Households do not observe the common urban income component z_1 directly. Their prior belief is

$$z_1 \sim \mathcal{N}(\bar{z}, \sigma^2_z), \qquad \bar{z} > 0$$

Because urban incomes share the common component z_1, migration outcomes are corre-

[4] Assumption 2 below imposes conditions ensuring that the certainty equivalent of rural income is positive.

[5] The model represents informational and financial relationships through a single undirected, unweighted network. A link ij denotes an underlying relationship through which households may exchange migration information, share resources, or both; direction-specific exchanges such as borrowing/lending or receiving/providing advice are therefore represented by the same link. Links are unweighted because the model records only whether the relationship exists, not its intensity.

lated across households:

$$\text{Corr}(e_{i1}, e_{j1}) = \frac{\sigma_z^2}{\sigma_z^2 + \sigma_{\tilde{e}}^2}, \qquad i \neq j.$$

Moreover,

$$\text{Cov}(z_1, z_2) = \rho\sigma_z^2,$$

so observing a period-1 urban income realization provides information about urban returns in period 2.

The network determines which migration outcomes households observe. Migration choices are publicly observable, but households directly observe the realized urban incomes of their pre-existing neighbors only when those neighbors migrate. Let $\mathcal{I}_{i1}$ denote household i's information after period $t = 1$ outcomes have been realized. In particular,

$$\mathcal{I}_{i1} = \sigma\left(G_1, \{M_{k1}\}_k, y_{i1}, \{y_{j1}\}_{j\in N_{i1}}, e_{i1}M_{i1}, \{e_{j1}M_{j1}\}_{j\in N_{i1}}\right).$$

Thus, households with more migration exposure in their existing networks have more information about the common component of urban returns.

Income Sharing. Households share income with their neighbors. In line with Bramoullé and Kranton (2007), Ambrus and Elliott (2021), and Ambrus et al. (2022), we assume that the network structure determines the income-sharing arrangement.[6] Under the sharing rule, household i allocates an equal share of its rural and urban income to itself and each of its d_{it} neighbors in each period $t \in \{1, 2\}$. The net bilateral transfer from household i to household $j \in N_{it}$ at period t is therefore

$$t_{ij,t} = \frac{y_{it} + e_{it}M_{it}}{d_{it} + 1} - \frac{y_{jt} + e_{jt}M_{jt}}{d_{jt} + 1}.$$

Household i's consumption is

$$c_{it} = y_{it} + (e_{it} - q)M_{it} - \sum_{j\in N_{it}} t_{ij,t}.$$

[6]We abstract from local information and contracting constraints such as limited observability or verifiability of income realizations. Such constraints could reduce the amount of income that can effectively be shared and could be incorporated by introducing a scaling parameter into the transfer rule. They are omitted here to keep the analysis tractable and separate their effects from the main mechanism.

Endogenous Link Formation and Learning. Between the two migration decisions, households may form new bilateral links at cost $\kappa > 0$ per link, measured in consumption units. If i forms (does not form) a link with any household j we have $L_{ij,2} = 1$ ($L_{ij,2} = 0$). Households can form links with any number of households in their XXX. Existing links cannot be severed, so the endogenous network G_2 contains all links in G_1 as well as any newly formed relationships. Section 2.2 specifies the conditions under which new links forms.

New links allow households to acquire information about period-1 migration outcomes that was unavailable through the initial network. In particular, when household i forms a new link with household j and j migrated in period $t = 1$, household i receives the signal about urban income from period $t = 1$

$$s_{i1}^{j} = e_{j1} + \epsilon_{j1},$$

where

$$\epsilon_{j1} \sim \mathcal{N}\left(0, \frac{\sigma_{\epsilon}^{2}}{1 + d_{j1}^{M}}\right).$$

Here d_{j1}^{M} denotes the number of j's period-1 neighbors who migrated. The underlying idea is that households with greater migration exposure have observed more realizations of urban income and can therefore transmit more precise information about migration opportunities. A household that did not migrate in period $t = 1$ provides no direct signal about its own urban income realization.

Pre-existing and newly formed links therefore differ in how information is obtained. Household i directly observes the realized migration incomes of its pre-existing neighbors, so the only noise contained in these signals is the idiosyncratic earnings term $e_{j1} = z_1 + \tilde{e}_{j1}$. Information acquired retrospectively through a new link is additionally subject to communication noise.[7]

After new links are formed and their signals are transmitted, household i's information set for its period-2 migration decision is

$$\mathcal{I}_{i,2} = \sigma\left(\mathcal{I}_{i1}, G_2, \{s_{i1}^{j}\}_{j \in N_{i,2} \setminus N_{i1}}\right).$$

Since existing links cannot be severed, $\mathcal{I}_{i,2}$ is a weak superset of the period-1 information set $\mathcal{I}_{i1}$.

[7] If signals are transmitted perfectly, so that there is no communication noise, the main results remain unchanged: a new link still provides an additional observation of migration returns and therefore reduces uncertainty. However, migration exposure among household j's contacts no longer affects the precision of the information transmitted by j.

Let N_{i2} denote household i's set of neighbors in G_2, and let $\mathcal{L}_i \equiv N_{i2} \setminus N_{i1}$ denote the set of new neighbors formed by household i through the link-formation stage. Hence, $n_i \equiv |\mathcal{L}_i| = d_{i2} - d_{i1}$.

To rule out contemporaneous informational spillovers across new links, we make the following simplifying assumption:

Assumption 1. *New-link formation depends only on information available prior to the link-formation stage. Information obtained through newly formed links does not affect the formation of other links during that stage.*

That is, households evaluate potential relationships using $\mathcal{I}_{i1}$; signals acquired through those relationships become available only after the new network has been formed and therefore affect subsequent migration rather than other link-formation decisions in the same stage.

In summary, the effect of a new connection can be separated into changes in expected consumption and changes in consumption variance. A new connection reduces the share of household i's own income that it retains, while allowing it to receive a share of household j's income. As j's degree grows in comparison to i's, the connection becomes less attractive because the net amount transferred from j to i falls as it is spread across more neighbors.

A new link reduces i's exposure to its own income risk, while also exposing i to fluctuations in j's income. This additional exposure is weaker when j has a larger degree, since the share of j's income accruing to i is smaller. At the same time, the information transmitted through linked households that have migrated reduces uncertainty about migration income. Thus, the net variance effect depends on whether the reduction in own-income and migration uncertainty dominates the additional exposure to the linked household's income risk.

We make two additional assumptions to XXX ...

Assumption 2. *A positive rural income certainty equivalent implies $CE(y_{it}/(d_{it}+1)) > 0$ for every $d_{it} \geq 0$, since expected income decreases linearly, whereas its variance decreases quadratically, with $1/(d_{it}+1)$.*[8]

The condition for this to be true, for any d_{it} is

$$\mathbb{E}[y_{it}] > \frac{r}{2} Var(y_{it}), \qquad \forall i, t.$$

This condition guarantees that the certainty equivalent of the household's share of rural in-

[8]Throughout the paper, CE denotes the certainty equivalent under CARA preferences. Its argument specifies the payoff being evaluated. In particular, $CE(X)$ denotes the certainty equivalent of a given income lottery X, while $CE_i(G)$ denotes household i's ex-ante continuation certainty equivalent under network G, accounting for subsequent optimal migration decisions.

come is positive for any degree $d_{it} \geq 0$:

$$CE\left(\frac{y_{it}}{d_{it}+1}\right) = \frac{\mathbb{E}[y_{it}]}{d_{it}+1} - \frac{r}{2}\frac{Var(y_{it})}{(d_{it}+1)^2} > 0, \qquad \forall d_{it} \geq 0.$$

Assumption 3 (Positive Prior Urban-Income Certainty Equivalent)**.** *Under the prior information, the certainty equivalent of the household's share of urban income is positive for any degree. In particular,*

$$\mathbb{E}[e_{it}] > \frac{r}{2}Var(e_{it}), \qquad \forall i, t.$$

This condition guarantees that

$$CE\left(\frac{e_{it}}{d_{it}+1}\right) = \frac{\mathbb{E}[e_{it}]}{d_{it}+1} - \frac{r}{2}\frac{Var(e_{it})}{(d_{it}+1)^2} > 0, \qquad \forall d_{it} \geq 0.$$

2.2 Timing of the Model

The model has two periods ($t = 1$ and $t = 2$) and two migration decisions (one per period) separated by an endogenous network-formation stage.

At the beginning of period $t = 1$, the initial network G_1 is exogenously given. Households simultaneously choose whether to migrate, with $M_{i1} \in \{0, 1\}$. Migration choices are publicly observable. Rural and urban incomes are then realized, and households observe the realized migration incomes of their pre-existing neighbors.

At the beginning of period $t = 2$, households may form new bilateral links. Existing links cannot be severed. For any pair of previously unconnected households i and j, a new link forms whenever it is mutually beneficial net of the link-formation cost κ. Let

$$\Delta CE_i(ij \mid G) = CE_i(G + ij) - CE_i(G) - \kappa$$

denote the net marginal value of link ij to household i, where G is a candidate period-2 network containing the inherited network G_1 and all other link-formation decisions taken as given, but excluding the potential link ij. Hence, $G_1 \subseteq G$ and $ij \notin G$. This differs from G_2, which denotes the resulting network after the link-formation stage has been completed. Since existing links cannot be removed, G_2 contains all links in G_1 together with the new links formed during this stage.

Because the value of a link depends on the other new links formed during the same stage, each potential link is evaluated conditional on the resulting network excluding that link (by keeping the other links fixed). We characterize a period-2 network G_2 satisfying the following bilateral conditions. Every included new link ij is mutually beneficial given the other links in the network:

$$\Delta CE_i(ij \mid G) \geq 0, \qquad \Delta CE_j(ij \mid G) \geq 0,$$

with at least one household strictly benefiting. Conversely, no excluded link is mutually beneficial given the resulting network.

Because $\Delta CE_i(ij \mid G)$ may itself depend on the other links contained in G, these bilateral conditions do not by themselves guarantee existence or uniqueness of a pure network satisfying them for every parameter configuration. We therefore state the theoretical results conditional on a period-2 network satisfying these conditions. If several such networks exist, the model does not impose an additional network-selection rule. [9]

Link formation depends only on information available prior to the network-formation stage. In particular, households cannot use information obtained through a newly formed link to determine which other links are formed during the same stage.

Once the new network G_2 has been formed in $t = 2$, newly connected households exchange information about period-1 migration outcomes. If household j migrated in period $t = 1$, a newly linked household i observes the signal s_{i1}^{j} defined above. Households update their beliefs using this newly available information and then choose whether to migrate in period $t = 2$, with $M_{i,2} \in \{0, 1\}$. Period-2 incomes and transfers are then realized and the game ends.

The timing can therefore be summarized as

$$G_1 \longrightarrow M_1 \longrightarrow \text{ period } t = 1 \text{ income realizations } \longrightarrow G_2 \longrightarrow \text{ new signals } \longrightarrow M_2.$$

where M_1 are migration choices in the economy at $t = 1$ and M_2 are migration choices in the economy at $t = 2$.

[9] Notice that the period-1 network is taken as given and is interpreted as an inherited risk-sharing network formed before the beginning of the model. Its existence therefore does not require every individual link to be optimal under the period-1 realization. If individual rationality is imposed ex ante, it requires only that each household's total expected certainty equivalent under the initial network be weakly greater than under isolation.

Payoffs. Households maximize expected intertemporal utility:

$$\mathbb{E}\left[u(c_{i,1}) + \beta u(c_{i,2})\right],$$

where $\beta \in (0,1)$ is the discount factor.

2.3 Migration Equilibrium

We characterize migration decisions by backward induction, conditional on a period-2 network satisfying the bilateral-consent conditions in Section 2.2. The analysis does not require uniqueness of this network; when multiple such networks exist, the results are understood conditional on the selected network.

Final-period migration. In period $t = 2$, migration no longer generates information useful for a subsequent migration decision. Conditional on the migration choices of the other households, household i migrates if and only if

$$\phi_{i,2} \equiv \frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i2}]}{d_{i2}+1} - q - \frac{r}{2}\left[\frac{\mathrm{Var}(e_{i2} \mid \mathcal{I}_{i2})}{(d_{i2}+1)^2} + \frac{2\ \mathrm{Var}(z_2 \mid \mathcal{I}_{i2})}{d_{i2}+1}\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1}\right] > 0. \tag{1}$$

The second variance term captures the additional covariance exposure created when household i migrates while already receiving migration income from migrating neighbors. If no neighbor migrates in period $t = 2$, this term is zero and the condition reduces to the standard certainty-equivalent comparison. In case of indifference, $\phi_{i,2} = 0$, either action is optimal. Derivations to obtain the cutoff condition for migration at $t = 2$ are provided in the Appendix.

Corollary 1 (Signals and Migration)**.** *The migration propensity ϕ_{i2} is increasing in favorable signal realizations received from neighbors.*

Under Gaussian updating, favorable signals raise the posterior expectation of urban income without increasing posterior variance. They therefore raise the certainty equivalent of migration and make migration more attractive.

As a benchmark, in the absence of informative signals, migration depends only on the prior distribution of urban returns:

$$\mathbb{E}[e_{i,2}] - \frac{r}{2}Var(e_{i,2}) > q.$$

In the random-walk special case $\rho = 1$, this condition becomes

$$\bar{z} - \frac{r}{2}\left(\sigma_\lambda^2 + \sigma_{\tilde{e}}^2 + \sigma_z^2\right) > q.$$

Early migration as experimentation. Migration in initial period $t = 1$ differs from migration in the final period because it also generates information that can be used in period $t = 2$.

Household i migrates at $t = 1$ if and only if

$$\phi_{i,1} \equiv \Delta\mathbb{E}[u(c_{i1})] + \beta\Delta\mathbb{E}[u(c_{i2})] > 0.$$

where Δ denotes the difference between migrating and not migrating in period $t = 1$. The first term captures the contemporaneous payoff from migration, while the second captures the continuation value of the information revealed by the household's own migration outcome. An explicit expression for $\phi_{i,1}$ is provided in the Appendix.

Proposition 1 (Strategic Substitutability in Current Migration)**.** *Under the conditions stated in the Appendix, period-*1 *migration decisions are strategic substitutes: household i's incentive to migrate weakly decreases when any neighbor switches from not migrating to migrating.*

If neighbors are symmetric, this implies a cutoff in the number of migrating neighbors. With heterogeneous neighbors, the strategic-substitution result continues to hold pairwise, but household i's migration decision depends on which neighbors migrate, rather than only on their number.

When few neighbors migrate, household i expects to receive little information about urban opportunities through its network and therefore has a stronger incentive to experiment by migrating itself. As more neighbors migrate, their realized outcomes provide additional information about urban returns, reducing the informational value of household i's own migration. Household i can therefore free-ride on the information produced by its migrating neighbors.

In case of indifference, $\phi_{i,1} = 0$, both migrating and not migrating are best responses. Hence, absent a tie-breaking rule, household i may choose either action, and multiple Nash equilibria may arise.

This strategic substitutability can also generate multiple period-1 migration equilibria under intermediate parameter values. Different households may be able to perform the role of early information producer, while their neighbors optimally wait and learn from their experiences. The formal conditions for multiplicity are provided in the Appendix.

Income and network comparative statics. We derive the following proposition about the relative attractiveness of rural and urban income streams:

Proposition 2 (Rural and Urban Income)**.** *Under the conditions stated in the Appendix, the migration propensity $\phi_{i,1}$ is decreasing in the certainty equivalent of rural income and increasing in the certainty equivalent of migration income:*

$$\frac{\partial \phi_{i,1}}{\partial CE(y_{i1})} < 0, \qquad \frac{\partial \phi_{i,1}}{\partial CE(e_{i1})} > 0.$$

An improvement in the certainty equivalent of rural income raises the value of remaining in the village and therefore discourages migration. Conversely, an increase in the certainty equivalent of urban income raises the return to migrating. Thus, both expected income and income risk matter for migration through their effects on the relative certainty equivalents of the two options.

This result also clarifies how risk-sharing connections can affect migration. Consider an isolated household and an otherwise comparable connected household facing the same information about migration returns. In particular, suppose that the connected household's neighbors do not migrate in period $t = 1$, so that the two households face the same uncertainty about urban income.

Corollary 2 (Network Isolation and Migration)**.** *Consider an isolated household and an otherwise comparable household in a nonempty initial network G_1. Suppose network participation raises household i's period-1 expected utility by at least as much when it does not migrate as when it migrates, so that*

$$\begin{aligned} &u\left(CE\left(y_{i1} - \sum_{j \in N_{i1}} t_{ij,1}\Big|_{M_{i1}=0}\right)\right) - u(CE(y_{i1})) \\ &\quad \geq u\left(CE\left(y_{i1} + e_{i1} - q - \sum_{j \in N_{i1}} t_{ij,1}\Big|_{M_{i1}=1}\right)\right) - u(CE(y_{i1} + e_{i1} - q))\,. \end{aligned}$$

Suppose also that information available through the initial network weakly reduces the marginal continuation value of household i's own period-1 migration, so that

$$\mathbb{E}[u(c_{i2}) \mid M_{i1} = 1, \varnothing] - \mathbb{E}[u(c_{i2}) \mid M_{i1} = 0, \varnothing] \geq \mathbb{E}[u(c_{i2}) \mid M_{i1} = 1, G_1] - \mathbb{E}[u(c_{i2}) \mid M_{i1} = 0, G_1],$$

where the period-2 continuation values are evaluated under the optimal period-2 migration decision after the available information is observed.

Then an isolated household has a weakly stronger incentive to migrate in period $t = 1$ than an otherwise comparable connected household: $\phi_{i,1}(\varnothing) \geq \phi_{i,1}(G_1)$. The inequality is strict whenever either of the two conditions above is strict.

Holding information fixed, the result follows from the insurance value of the network. If transfers raise the connected household's certainty equivalent, network participation improves its outside option relative to migration. An isolated household does not receive this benefit and consequently has a stronger incentive to migrate. Proofs are provided in the Appendix.

2.4 Link Formation Without Transfers

We first consider a benchmark in which newly formed links transmit information but do not affect income-sharing arrangements. This isolates the informational motive for network formation. Section 2.5 returns to the full model, in which a new link also creates risk-sharing and redistribution obligations.

Consider a potential link between households i and j. Because the signal provided by the new link is observed before the period-2 migration decision, household i evaluates the link anticipating that it can condition its subsequent migration choice on the information received. Define the gross informational value of the link as

$$\omega_{ij} \equiv -\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r \max_{M_{i,2}\in\{0,1\}} CE_i\left(M_{i,2} \mid \mathcal{I}_{i1}, s_{i1}^j\right)\right\} \,\middle|\, \mathcal{I}_{i1}\right] - \max_{M_{i,2}\in\{0,1\}} CE_i\left(M_{i,2} \mid \mathcal{I}_{i1}\right).$$

Thus, ω_{ij} is the maximum link-formation cost that household i is willing to incur to obtain the information provided by household j.

The signal reduces uncertainty about future urban income. Let

$$V_i \equiv \mathrm{Var}\left(e_{i,2} \mid \mathcal{I}_{i1}\right)$$

denote household i's conditional variance before forming the link, and let

$$V_i(j) \equiv \mathrm{Var}\left(e_{i,2} \mid \mathcal{I}_{i1}, s_{i1}^j\right)$$

denote its conditional variance after observing the signal provided by j. A more informative signal therefore implies a larger reduction in posterior uncertainty $V_i - V_i(j) \geq 0$.

Importantly, however, this reduction in posterior variance is not a separate additive component of the ex-ante value of information. If the period-2 migration decision were held fixed, the lower conditional variance after observing the signal would be exactly offset ex

ante by the dispersion of the posterior mean across possible signal realizations. By the law of iterated expectations, the signal therefore has no ex-ante value when the household cannot adjust its action in response to the information.

The informational value of the link instead arises from the possibility that the signal changes household i's optimal period-2 migration decision. Let $\mathcal{O}_i(ij) \geq 0$ denote this option value of adjusting the migration decision after observing the signal.[10] In the information-only benchmark, the entire informational value of the link is therefore its option value $\omega_{ij} = \mathcal{O}_i(ij)$.

Thus, although a more informative link reduces posterior uncertainty, the welfare gain from that information arises only because household i can condition its subsequent migration decision on the signal. A formal derivation of the cancellation under a fixed migration decision and of the option value is provided in the Appendix.

Household i's net gain from forming the link is $\omega_{ij} - \kappa$, where $\kappa > 0$ is the link-formation cost.

Proposition 3 (Information-Only Link Formation)**.** *A new informational link between households i and j forms if and only if*

$$\omega_{ij} \geq \kappa, \qquad \omega_{ji} \geq \kappa, \qquad \textit{and} \qquad \max\{\omega_{ij} - \kappa, \omega_{ji} - \kappa\} > 0.$$

Thus, in the absence of transfers, households form a link when the informational value of the relationship weakly exceeds its cost for both households and strictly exceeds it for at least one of them. This value depends on the uncertainty each household initially faces, the precision of the potential partner's signal, and, crucially, the possibility that the information changes the subsequent migration decision.

Corollary 3 (Uncertainty and Information Link Formation)**.** *Holding household i's prior information and link costs fixed, the value of a new information link is weakly increasing in the precision of the signal provided by household j. The effect of greater prior uncertainty about urban returns depends on household i's baseline migration decision and is therefore not unconditionally signed.*

The signal-precision result follows because a more precise signal provides weakly more information about future urban returns. Since household i observes the signal before choosing whether to migrate in period $t = 2$, it can always reproduce the decision it would have taken under a less precise signal. A more precise signal therefore weakly increases the value of the

[10] $\mathcal{O}_i(ij) = 0$ if the signal never changes the optimal migration decision in period $t = 2$.

information, with a strict increase whenever the additional precision changes the migration decision with positive probability.

In the signal structure described in Section 2.1, conditional on household j having migration experience to transmit, greater migration exposure among j's neighbors reduces the noise in the signal transmitted by j. Household j therefore becomes a more informative potential connection.

The effect of household i's prior uncertainty is more subtle. Greater uncertainty increases the amount of uncertainty that a signal can potentially resolve, but the value of information also depends on how the signal affects the subsequent migration decision. As shown in the Appendix, the effect of greater prior uncertainty on ω_{ij} therefore depends on household i's baseline migration choice and might be non-positive in all cases.

Information obtained from different links may also be partially redundant. Consider the information available to household i from its new information links. Under Gaussian updating, observing an additional informative signal from household j weakly reduces the posterior uncertainty about e_{i2}:

$$Var\left(e_{i2} \mid \mathcal{I}_{i1}, \{s_{i1}^k\}_{k \in N_{i2} \setminus N_{i1}}\right) - Var\left(e_{i2} \mid \mathcal{I}_{i1}, \{s_{i1}^k\}_{k \in N_{i2} \setminus N_{i1}}, s_{i1}^j\right) \geq 0.$$

Moreover, as household i accumulates informative signals, the posterior variance reduction generated by an additional signal becomes smaller.

This statistical diminishing-return property does not by itself imply that the welfare value of an additional link decreases monotonically. In the information-only benchmark, the welfare value of information is its option value, $\omega_{ij} = \mathcal{O}_i(ij)$, and therefore depends not only on the reduction in posterior uncertainty but also on whether the additional information changes household i's optimal period-2 migration decision.

Proposition 4 (Diminishing Informational Returns and Optimal Degree)**.** *For signals of a given precision, the reduction in posterior uncertainty generated by an additional informative link decreases as household i accumulates information.*

Suppose, in addition, that the decision value of successive informative links is weakly decreasing as household i accumulates informative signals. Then household i forms additional information links only while their marginal informational value weakly exceeds the link cost κ, subject to the potential partner also preferring the link.

If the number of new information links formed by household i, n_i^*, is interior, the last link formed satisfies

$$\mathcal{O}_i(ij \mid G; r) \geq \kappa,$$

whereas the next additional informative link satisfies

$$\mathcal{O}_i(ij \mid G; r) < \kappa.$$

The intuition is that information from different connections can be redundant. Under Gaussian updating, an early informative connection can reduce household i's uncertainty about urban returns, whereas an additional signal of a given precision produces a smaller reduction in posterior variance once several informative signals have already been observed. Corollary 3 shows that, holding pre-link beliefs fixed, the informational value of a connection is increasing in the amount of uncertainty it resolves, $V_i - V_i(j)$. Thus, the diminishing reduction in posterior uncertainty provides the statistical basis for diminishing informational returns from successive connections.

This statistical result does not by itself imply that $\mathcal{O}_i(ij \mid G; r)$ must decrease with every additional signal. The option value depends on whether the new information changes the subsequent migration decision, and previous signals can move household i closer to the migration threshold. The proposition therefore additionally requires that the diminishing statistical contribution of successive signals translates into a weakly diminishing marginal decision value. Under this condition, positive link costs prevent households from forming arbitrarily many information-seeking relationships. Depending on parameter values, the outcome may involve no new links, an interior number of links, or all available informative links.

This benchmark also highlights the role played by transfers in the full model. As shown in Corollary 3, greater signal precision weakly raises the informational value of a connection, while the effect of greater prior uncertainty depends on the household's migration incentives. Once links also create income-sharing relationships, these informational gains must additionally be weighed against the insurance benefits and redistribution costs generated by the same connection.

2.5 Link Formation With Transfers

We now return to the full model, in which newly formed links generate both information and income-sharing obligations. Relative to the information-only benchmark, a new connection can therefore affect household i through three channels: it provides information about urban returns, it can reduce consumption risk through income sharing, and it can redistribute expected income between the two households.

Because household i observes the additional signal before choosing whether to migrate in period 2, it can condition its migration decision on that signal. The value of a new

link therefore also includes an option value: the gain from adjusting the period-2 migration decision after observing the additional information, relative to following the migration policy that would have been optimal without the new link.

The net gain to household i from adding link ij to network G is

$$\Delta CE_i(ij \mid G) = CE_i(G + ij) - CE_i(G) - \kappa,$$

where ${}_i(G+ij)$ and $CE_i(G)$ denote the ex-ante continuation certainty equivalents under the two networks, taking into account that the period-2 migration decision is chosen optimally after the information available under each network is observed.

The gain from the link is decomposed as follows

$$\Delta CE_i(ij \mid G) = CE_i(G + ij; M_{i,2}^G) - CE_i(G; M_{i,2}^G) + \mathcal{O}_i(ij \mid G) - \kappa,$$

where the first two terms capture the financial effect of the new income-sharing relationship, holding fixed the migration policy that would have been optimal under network G. This effect shows changes in expected consumption through redistribution and changes in consumption risk through insurance or additional exposure to transfers. This is evaluated by using CARA certainty equivalents.

Notice that the reduction in posterior uncertainty generated by the additional signal is not a separate variance-reduction benefit in the expression. As in the information-only benchmark, holding the migration decision fixed, the reduction in posterior variance is outweighed ex ante by the dispersion of posterior means across signal realizations. The informational value of the link instead arises through the possibility of adjusting the subsequent migration decision after observing the signal.

The term $\mathcal{O}_i(ij \mid G) \geq 0$ is the option value of adapting the period-2 migration decision after observing the additional signal. Contrary to the information-only benchmark, the total value of the link also contains the direct effects of the new income-sharing relationship on expected consumption and consumption risk. Formal expressions for these components are provided in the Appendix.

Proposition 5 (Trade-Off in Link Formation)**.** *Taking all other households' link-formation decisions as given, a new link between households i and j forms if and only if it is mutually beneficial.*[11]

$$\Delta CE_i(ij \mid G) \geq 0, \qquad \Delta CE_j(ij \mid G) \geq 0, \qquad \textit{and} \qquad \max\{\Delta CE_i(ij \mid G), \Delta CE_j(ij \mid G)\} > 0.$$

[11] We adopt the tie-breaking convention that an indifferent household accepts the link if the other household strictly benefits. If both households are indifferent, we have multiplicity of equilibrium.

The value of the link reflects the informational value of adapting the subsequent migration decision, the gains from risk sharing, and any expected redistribution generated by the new relationship, net of the link formation cost.

This proposition captures the central trade-off of the model. As in the information-only benchmark, a link is more valuable when it provides information that substantially reduces uncertainty about urban returns. Income sharing creates an additional insurance benefit when the new relationship reduces consumption risk. At the same time, a household that expects to transfer resources through the new relationship experiences a reduction in expected consumption. Holding the informational and insurance benefits fixed, larger expected redistribution obligations therefore make link formation less attractive.

The remainder of this subsection derives implications from this decomposition. We first show how income sharing creates strategic interaction among links to the same potential partner. We then consider how link value varies with partner characteristics, the number of existing links, and household risk aversion.

Proposition 6 (Strategic Substitutability in Link Formation)**.** *Consider two households, i and k, that might each form a new link with the same potential partner j. Holding fixed household j's period-1 migration outcome and migration exposure, link kj does not change the precision of the period-1 information transmitted by j to household i. However, it increases j's degree and therefore change the income-sharing value of link ij. Also, in the full model, this can also affect the option value associated with household i's subsequent migration decision.*

The difference in the marginal value of link ij is

$$\begin{aligned}&\Delta CE_i(ij \mid G + kj) - \Delta CE_i(ij \mid G)\\ &= [\Delta \mathbb{E}[c_{i,2} \mid G + kj] - \Delta \mathbb{E}[c_{i,2} \mid G]] - \frac{r}{2}[\Delta Var(c_{i,2} \mid G + kj) - \Delta Var(c_{i,2} \mid G)] + [\mathcal{O}_i(ij \mid G + kj) - \mathcal{O}_i(ij \mid G)].\end{aligned}$$

If the reduction in household i's expected share of j's income dominates the accompanying reduction in consumption risk and any offsetting increase in the option value, then

$$\Delta CE_i(ij \mid G + kj) < \Delta CE_i(ij \mid G).$$

Under this condition, potential links from different households to the same partner are strategic substitutes.

The result follows from the income-sharing rule. When household k forms a link with household j, j's degree increases and its income is divided across a larger number of relationships. This reduces the expected income-sharing benefit that household i obtains from

subsequently linking to j. The increase in j's degree also reduces household i's exposure to variation in j's income, partially offsetting the decline in expected income. Under the condition in the proposition, the expected-income effect dominates, so the marginal value of link ij falls when link kj is present.

This strategic-substitutability mechanism is absent from the information-only benchmark. The information transmitted by household j concerns migration outcomes realized in period $t = 1$, and its precision depends on j's period-1 migration exposure. Both are determined before new links form. Consequently, link kj does not change the informational value to household i of link ij. Because information-only links also create no income-sharing obligations, the value of link ij is independent of other new links to household j. The proof is provided in the Appendix.

Corollary 4 (Characteristics of Attractive Partners)**.** *Holding other components of link value fixed, the incentive to link is decreasing in the household's expected net redistribution to the potential partner and increasing in the precision of the information that partner provides.*

Conditional on household j having migration experience to transmit, greater migration exposure among j's neighbors raises the value of linking to j by increasing signal precision. Under the income-sharing rule of Section 2.1 and the conditions stated in the Appendix, the value of linking to j is decreasing in j's total degree, holding migration exposure fixed.

The two dimensions of a potential partner's network therefore play distinct roles. A household connected to more migrants can be a more valuable information provider because it transmits a more precise signal about urban opportunities. By contrast, a household with a larger total network divides its income among more relationships. Under our sharing rule, this reduces the value of the income-sharing relationship to a new partner. Thus, all else equal, households prefer information providers with greater migration exposure but fewer competing income-sharing relationships.

Corollary 5 (Marginal Value of Additional Links)**.** *Under the diminishing marginal decision-value condition in Proposition 4, the marginal informational value of an additional link declines as household i accumulates informative connections. The income-sharing component of marginal link value, however, may be positive or negative. Consequently, introducing transfers can either increase or decrease the number of links formed relative to the information-only benchmark.*

The diminishing returns to information established in Section 2.4 continue to apply here: once a household has already obtained several informative signals, another signal contributes relatively little. The income-sharing component does not have an unambiguous effect in the

same direction. An additional relationship may improve diversification and make another link more attractive, or it can impose large redistribution costs to make the link unattractive. Therefore, the full-model network can be either denser or sparser than the information-only network.

Corollary 6 (Risk Aversion and Link Formation)**.** *The effect of greater risk aversion on the value of a new link is generally ambiguous in the full model. Holding fixed the migration policy that would be optimal under G,*

Holding fixed the migration policy that would be optimal under G, the full-model derivative is

$$\frac{\partial \Delta CE_i(ij \mid G)}{\partial r} = \frac{\partial}{\partial r}\left[CE_i(G+ij; M_{i,2}^G) - CE_i(G; M_{i,2}^G)\right] + \frac{\partial \mathcal{O}_i(ij \mid G; r)}{\partial r}.$$

Under fixed migration actions and a fixed continuation migration profile, the financial derivative reduces to $-\frac{1}{2}\Delta \operatorname{Var}(c_{i2})$.

Then, the first term captures how the new income-sharing relationship changes consumption risk, while the second captures how risk aversion changes the value of adapting the subsequent migration decision after receiving the additional information.

In the information-only benchmark, $\omega_{ij} = \mathcal{O}_i(ij \mid G; r)$. Locally within a region in which the baseline migration choice remains unchanged,

$$M_{i,2}^G = 0 \quad \Longrightarrow \quad \frac{\partial \omega_{ij}}{\partial r} \leq 0,$$

whereas when $M_{i,2}^G = 1$ the effect of risk aversion on the informational value of the link is generally ambiguous.

When $M_{i,2}^G = 0$, information is valuable only when sufficiently favorable signals induce the household to migrate. Greater risk aversion makes these risky migration opportunities less valuable and therefore weakly reduces the value of the information. When $M_{i,2}^G = 1$, whereas information can allow the household to avoid migration after sufficiently unfavorable signals. Greater risk aversion changes both the value of this option and the value of the baseline migration action, so the overall informational effect is not generally signed.

With income sharing, the overall effect is generally ambiguous even conditional on the baseline migration choice. If the new relationship reduces consumption variance, $\Delta Var(c_{i,2}) < 0$, greater risk aversion increases the insurance value of the link. If instead it increases consumption variance, $\Delta Var(c_{i,2}) > 0$, greater risk aversion makes the relationship less attractive. In addition, the full-model option value might not inherit the information-only sign because the additional signal can affect the value of the income-sharing environment itself.

In the information-only benchmark, greater risk aversion does not operate through a separate posterior-variance-reduction benefit. The value of information is entirely the option value generated by the possibility of adapting the subsequent migration decision. When the household would not migrate without the additional signal, greater risk aversion makes favorable migration opportunities harder to exploit and weakly reduces the value of the link. When the household would otherwise migrate, information can instead be valuable because it identifies sufficiently unfavorable states in which migration should be avoided, and the overall effect of risk aversion is generally ambiguous.

In the full model, the same informational option-value channel remains, while income sharing introduces an additional effect through consumption risk. A new relationship can reduce consumption variance and provide insurance, in which case greater risk aversion increases the value of this component of the link. Alternatively, the relationship exposes the household to additional variation in transfers and increase consumption variance, in which case greater risk aversion makes the link less attractive. Hence, the relationship between risk aversion and network formation depends on the interaction between the informational option value and the change in consumption risk generated by the new relationship.

3 Data and Experiment

We test the model's predictions using data collected as part of the large-scale migration experiment studied in Barnett-Howell et al. (2026).[12] The study followed approximately 16,800 households in 560 rural villages across five Kenyan counties between 2022 and 2024. The data combine a short census of all households in each village with a more detailed panel survey of approximately 30 households per village. The panel data include information on named social-network measures, beliefs about potential urban incomes and migration risk, detailed information on migration histories and outcomes, economic outcomes in the village and for migrants, and information on financial transfers within the village. The study also included randomized interventions providing households with information about conditions in urban areas; we use this exogenous shock to households' information sets to test some of our model's predictions.

[12]See Barnett-Howell et al. (2026) for a fuller description of the sampling design, intervention implementation, and primary experimental results.

3.1 Sample and Data Collection

Villages and households were selected from the rural population using a three-stage sampling procedure. The study first randomly selected 560 sub-locations and then one village within each selected sub-location. With the assistance of local leaders, the research team constructed a census of approximately 53,000 households across these villages. Roughly 30 households per village were then randomly selected for the experimental panel. Households expressing some intention to send a migrant to a city were oversampled, and the empirical analysis uses sampling weights to adjust for this non-uniform selection and for survey non-response.

Data were collected in four waves. An initial household census was conducted between May and August 2022 and collected basic demographic information, migration histories and intentions, beliefs about urban earnings, and measures of social connections at the origin and destination. A longer baseline survey was administered between September and November 2022 and expanded the measurement of household income, economic activity, risk preferences, migration beliefs and uncertainty, and social-network relationships. A phone midline survey was conducted between May and August 2023, approximately eight months after treatment, and an in-person endline survey was conducted between February and May 2024, approximately sixteen months after treatment. The endline was completed directly with approximately 95% of the experimental sample.

The main surveys collected information about all relevant household members. During the baseline, midline, and endline rounds, the study also attempted separate phone interviews with other household members, focusing particularly on current and former urban migrants. These interviews provide direct reports of migrants' employment, earnings, and experiences at the destination. The panel therefore allows us to measure both household-level migration choices and the urban and rural income streams underlying those choices at several points in time.

3.2 Measuring Beliefs, Migration, and Social Networks

The surveys contain several measures corresponding closely to the objects in our model. Households report whether they intend to send a migrant, whether household members subsequently migrate, where migrants move, and the income earned by household members in rural and urban locations. Repeated income measurements allow us to study how migration responds to the level and variability of rural and urban income.

To measure beliefs about migration returns, households were asked about typical earnings in Nairobi and how much a particular member of their own household could earn after migrating. They were also asked for optimistic and pessimistic assessments of that potential

income. We use the gap between these assessments, relative to expected earnings, as a measure of subjective uncertainty about migration returns.

Of particular importance for this paper, the surveys collected named social-network data within each village. The informational network identifies households from whom the respondent would seek advice about finding a job in a nearby town or in Nairobi. The financial network identifies households the respondent would approach to borrow money. A third network identifies households that could provide help with farm activities such as plowing or harvesting. The surveys also measure contacts in Nairobi who could provide housing or job-search assistance and broader features of the village social environment, including trust, social interaction, and the prevalence of borrowing.

Later surveys provide repeated measures of these relationships and distinguish households that report receiving migration advice from those that report providing it. This distinction helps separate the demand for information from its supply: households with little knowledge of Nairobi may seek advice, while households with migration experience may be approached as information providers.

3.3 Design of the Randomized Information Shock

The parent experiment assigned villages to five arms that varied how information about the capital city, Nairobi, was provided to participants. All treatment arms included an information sheet and a script describing earnings, employment rates, rental costs, and living conditions in Nairobi. Certain arms facilitated matches between study households and more experienced migrants residing either in the rural village or in Nairobi: we exclude these households from treatment-effect regressions to retain a clean interpretation of the treatment as an information shock. Households in control villages received no intervention. We follow the estimating equations of Barnett-Howell et al. (2026) when estimating treatment effects, including controls for randomization strata, pre-experimental outcome values, and survey-timing controls.

Because the vast majority of households in the study initially underestimated Nairobi earnings, the information generally shifted expected returns upward. By providing information about typical, low, and high earnings, employment probabilities, and destination costs, it could also reduce perceived uncertainty. In the terminology of the model, the intervention may therefore both increase expected urban income and reduce its perceived variance:

$$\frac{\partial \phi_{i2}}{\partial \mathbb{E}[e_{i2} \mid s]} > 0 \qquad \text{and} \qquad \frac{\partial \phi_{i2}}{\partial Var(e_{i2} \mid s)} < 0,$$

where ϕ_{i2} denotes household i's migration propensity. Our model predicts that these external

shocks to households' information sets will act as substitutes for learning through endogenous link formation.

3.4 Mapping the Data to the Model

The theoretical network is represented as an undirected and unweighted graph. A link in the model should be understood as a high-level representation of the relationship between two households, summarizing the different interactions that the relationship can support. These interactions may in reality be specific and directed. For example, household i may seek migration information from household j, while household j may lend resources or provide farm assistance to household i. The model does not distinguish between these different functions or directions of the relationship, but instead represents them through a single link between the two households. Following a common approach in the empirical literature on rural networks, we take a household's village to define its set of potential network partners (Fafchamps and Gubert, 2007, Udry and Conley, 2004). This restriction is consistent with evidence that economically relevant social interactions in rural settings are highly localized. For example, Udry and Conley (2004) document that a majority of information-sharing and financial interactions in their study villages in Ghana occur within village boundaries.

The data allow us to characterize these relationships more precisely. In particular, we distinguish between informational, financial, and farm-assistance relationships and, when the survey permits, identify the direction of the interaction, such as giving or receiving advice and borrowing or lending. We use these more detailed measures when the distinction between the nature or direction of the relationship is relevant to the empirical analysis.

The data also distinguish between the availability and the active use of a relationship. Pre-treatment questions asking whom a household would approach measure the set of contacts available for information, credit, or assistance. Later reports of actual advice giving and receipt measure realized information flows. We use the former to characterize households' initial information and insurance opportunities and the latter to study changes in network use and formation.

Together, the data allows for three sets of tests. First, the pre-treatment cross-section allows us to relate information links to uncertainty, migration experience, risk preferences, and redistribution obligations. Second, the panel allows us to test how migration responds to urban returns, rural opportunities, and income risk. Third, the experiment allows us to test whether externally supplied information substitutes for network-based learning and whether it changes migration and network behavior differently for households with more or fewer pre-existing information connections.

3.5 Testable Implications of the Theoretical Model

Before turning to the empirical tests of our model in Section 4, we first enumerate the main predictions of the model that we bring to the data. We present them in the order in which they are tested.

H1: Redistribution obligations and link formation. Households are less likely to form information-seeking links when those relationships entail larger expected net transfers to the potential partner.

H2: Risk aversion and link formation. The effect of risk aversion on link formation is generally ambiguous and depends on both migration incentives and the risk created by the relationship.

For informational links, greater risk aversion reduces the value of information for households that would otherwise not migrate, but it is ambiguous for households that would otherwise migrate. When links also involve income sharing, greater risk aversion increases the value of links that improve insurance but decreases the value of links that create additional transfer risk. Hence, more risk-averse households are less likely to form links when the additional risk associated with the relationship dominates these informational and insurance benefits.

H3: Uncertainty and information links. Holding the household's expected migration return and the informativeness of the potential new link fixed, greater uncertainty about migration returns increases the value of an information link when migration is optimal before the additional signal is observed, $M_{i,2}^G = 1$. When $M_{i,2}^G = 0$, the effect of greater uncertainty is generally ambiguous.[13] This hypothesis is a prediction of the information-only benchmark.

H4: Incentives for advice provision and seeking. Households with migration experience are more valuable sources of information and are therefore more likely to be approached for migration-related advice. On the advice-seeking side, households for whom migration information is more relevant to their subsequent migration decision, or who expect greater net benefits from redistribution through social connections, have stronger incentives to seek advice.

[13]Here $M_{i,2}^G = 1$ means that migration is optimal before observing the additional signal. The result is therefore conditional on the household's baseline migration choice, not a comparison of link formation between households with $M_{i,2}^G = 1$ and $M_{i,2}^G = 0$. Information has value only when it can change the subsequent migration decision; for a household with $M_{i,2}^G = 1$, for example, a sufficiently unfavorable signal can induce it not to migrate.

H5: Income, risk, and migration. Households are more likely to migrate when expected urban returns are higher and less likely to migrate when rural opportunities are more attractive. Holding expected rural income fixed, greater rural income risk can also increase migration by reducing the certainty equivalent of remaining in the village.

H6: Externally provided information. An external information intervention has a larger effect on migration among households with fewer pre-existing sources of migration information. Given that externally provided and network-based information are substitutes when $M_{i,2}^G = 1$ in theory, and then the intervention reduces the marginal value of forming additional information-seeking links.

H7: Characteristics of potential partners. Among potential partners with migration experience, greater total connectedness in the baseline reduces their attractiveness for new link formation, holding migration exposure fixed, as characterized in Corollary 4.

4 Empirical Results

This section presents the results of our empirical analysis, guided by our hypotheses from Section 3.5.

4.1 Correlates of Information Connections

Table 1 presents the tests of our hypotheses related to information connections between households in a social network (village). All regressions control for the household's total number of social connections at baseline, so that coefficients can be interpreted as a difference in the number of information links holding total connectedness fixed. The top panel presents our basic specifications with no demographic controls. We find evidence consistent with hypothesis 1: our measure of financial reliance in the village (the share of households reporting that households in their village often borrow from one another) is strongly negatively correlated with information links: moving from 0 to 1 on this index reduces the average number of information links by about 27% ($p < 0.01$). This suggests that households view link formation as reflecting a trade-off between learning gains and redistribution costs: the more likely relationships are to entail redistribution obligations, the smaller the number of information links conditional on overall connectedness.

To test hypothesis 2, we examine whether households self-describe as unwilling to take risks. The information-only benchmark predicts that greater risk aversion reduces the value of information when the household would otherwise not migrate, while the effect is ambiguous

when it would otherwise migrate. We therefore interact risk aversion with a census-survey indicator for whether the household plans to send a migrant to Nairobi within the next year. Among households not planning to migrate, risk aversion is associated with about 9% fewer information links ($p = 0.02$). The interaction with planned migration is positive (coefficient $= 0.08$, $p = 0.13$), implying that this negative relationship is attenuated among households that might migrate. The same pattern remains after adding demographic controls. Although stated migration plans are only an imperfect empirical analogue of the model's baseline migration choice, the direction of the interaction is consistent with the information-only comparative static.

Finally, we examine hypothesis 3, which predicts in the information-only benchmark that greater uncertainty raises the value of an information link when the household would otherwise migrate, while the effect is ambiguous when it would not. We measure urban income uncertainty using households' optimistic and pessimistic assessments of potential Nairobi earnings, normalized by expected earnings. Because this hypothesis concerns subsequent information acquisition, Column 3 relates baseline uncertainty to the number of households with whom the respondent reports discussing migration at endline.[14] Among households not planning to migrate, a one-unit increase in baseline uncertainty is associated with about 14% more endline information links ($p = 0.07$). Contrary to the predicted differential, the interaction with planned migration is negative and statistically uncertain (coefficient $= -0.10$, $p = 0.40$). The implied association nevertheless remains positive for households that might migrate as well. Results are nearly unchanged with demographic controls. As above, stated migration plans are only an imperfect proxy for the model's baseline migration choice.

To help account for potential omitted variable bias in the relationships documented above, we control for household demographic characteristics in the lower panel of Table 1. In particular, we control for household size, the educational attainment of the highest-educated person in the household, total household expenditure, a binary variable indicating whether the household had a member working in Nairobi during the census (first) survey, a binary variable indicating whether the household planned to send a migrant to Nairobi as of the census survey, the household's prior belief about typical incomes in Nairobi as of the census survey, and the distance between the household and its county capital. These control variables affect our results very little.

[14] Using a later outcome helps separate baseline uncertainty from the information relationships that may subsequently respond to it.

4.2 Results on Advice Seeking

The results of Table 1 take observed social connections as an outcome of interest. However, these observed connections reflect the equilibrium result of both agents' behavior: for example, a household with good information about Nairobi will have less need for links, but will also be more desirable as a link from the perspective of other households. In such cases we refer to linking incentives as *asymmetric*. To analyze the role of asymmetric incentives, we decompose information links into those in which the household was primarily *receiving* advice and those in which it was *giving* advice.

Table 2 presents regression results with either the number of information connections the household gave advice to, or the number that it received advice from, as outcome variables. These outcomes are measured in the endline survey and reflect conversations the household reports having over the past year. We find that households with prior or current migrants in Nairobi—who are likely to be desirable as connections—are giving advice 25% more ($p < 0.01$), with no significant difference in advice receipt. This result is consistent with hypothesis 4: households with experience in Nairobi are especially likely to be approached by agents hoping to learn more about conditions in Nairobi.

On the advice seeker side, we consider two predictors of incentives to form links: intentions to migrate to Nairobi (which increases the informational value of learning about Nairobi), and a stated willingness to borrow from other households in the village (which increases the redistributive value of links). Both of these variables predict more advice receipt (about 20% more; p-values < 0.10) but have no significant relationship with advice giving. These results are consistent with households who have greater incentives to form links—either to learn about Nairobi or to increase their borrowing options—seeking out additional information connections. The results are consistent with hypothesis 4. In theory, if j did not migrate, the link provides no migration information and $\omega_{ij} = 0$; conditional on j having migrated, the signal can have positive decision value.

These results are only minimally changed when we include the same demographic control variables described in Section 4.1, as shown in the lower panel of Table 2.

Table 1: Correlates of Link Formation

	Baseline Potential Information Links		Endline Realized Information Links
No Controls			
Financial Reliance Index in Village	-0.27***		
	(0.07)		
	[0.00]		
Risk Averse		-0.09**	
		(0.04)	
		[0.02]	
Risk Averse × Might Migrate		0.08	
		(0.06)	
		[0.13]	
Urban Income Uncertainty			0.13*
			(0.07)
			[0.07]
Urban Income Uncertainty × Might Migrate			-0.10
			(0.12)
			[0.40]
+ HH Demographic Controls			
Financial Reliance Index in Village	-0.35***		
	(0.08)		
	[0.00]		
Risk Averse		-0.08**	
		(0.04)	
		[0.04]	
Risk Averse × Might Migrate		0.08	
		(0.06)	
		[0.16]	
Urban Income Uncertainty			0.13*
			(0.07)
			[0.07]
Urban Income Uncertainty × Might Migrate			-0.09
			(0.12)
			[0.48]
Dep. Var. Mean	0.63	0.62	0.16
Observations	16,349	16,291	14,895

All models use Poisson regression. Each observation is a household. In Columns (1)–(2), the dependent variable is the number of social connections the household reports it could use for information about Nairobi at baseline. In Column (3), the dependent variable is the number of information connections with whom the household reports discussing migration to Nairobi at endline. The latter therefore captures realized information exchange after baseline rather than the baseline stock of potential information sources. *Financial Reliance Index* is the share of households in a village who report that people there often rely on one another for informal borrowing. *Risk Averse* is a binary variable indicating whether the respondent does not consider themselves generally willing to take risks. *Urban Income Uncertainty* is the gap between the respondent's optimistic and pessimistic belief about their potential income in Nairobi, divided by their expected income. All regressions control for the total number of social connections (information + financial + farm assistance). Columns 2 and 3 include a main effect for pre-treatment planned migration and its interaction with the focal regressor; the main effect of planned migration is not reported. Standard errors clustered at the village level in column 1 and heteroskedasticity-robust in columns 2–3; two-sided p-values in brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Correlates of Advice Seeking and Giving

	Dep. Var.: # Connections Gave Advice To			Dep. Var.: # Connections Received Advice From		
No Controls						
Had Migrated to Nairobi	0.25***			0.06		
	(0.09)			(0.12)		
	[0.00]			[0.62]		
Plans to Migrate to Nairobi		0.05			0.22*	
		(0.09)			(0.12)	
		[0.54]			[0.06]	
Would Seek Borrowing From Village			-0.08			0.21*
			(0.09)			(0.13)
			[0.37]			[0.10]
+ HH Demographic Controls						
Had Migrated to Nairobi	0.23***			0.06		
	(0.09)			(0.12)		
	[0.01]			[0.61]		
Plans to Migrate to Nairobi		0.04			0.23*	
		(0.09)			(0.12)	
		[0.70]			[0.06]	
Would Seek Borrowing From Village			-0.10			0.23*
			(0.09)			(0.13)
			[0.26]			[0.07]
Dep. Var. Mean	0.13	0.13	0.13	0.04	0.04	0.04
Observations	15,988	15,988	15,988	15,988	15,988	15,988

All models use Poisson regression. Each observation is a household. Dependent variable is the number of social links the household reports giving advice about Nairobi to (columns 1–3) or receiving advice about Nairobi from (columns 4–6). All regressions control for the total number of social connections (information + financial + farm assistance). Heteroskedasticity-robust standard errors in parentheses; two-sided p-values in brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.3 Results on Migration

We proceed to test our hypotheses related to households' migration behavior in Table 3. Column 1 regresses the number of Nairobi migrants at endline on Nairobi income at baseline and all other income at baseline. The number of Nairobi migrants at baseline is included as a control (not shown in output) because otherwise there would be an omitted relationship between the number of migrants in Nairobi at endline and Nairobi income at endline, operating through the persistence of migrant counts over time. We find evidence consistent with hypothesis 5: greater urban income in period $t = 1$ is associated with more migration in period $t = 2$ (10% increase per one-standard-deviation change in urban income; $p < 0.01$), while greater rural income in period $t = 1$ is associated with less migration (6% decrease per one-standard-deviation change in rural income; $p = 0.02$). Note that the latter result does not support a simple prediction about binding credit constraints for a large fraction of households: in a credit-constraint model, additional income (regardless of its source) would predict more migration.

In column 2 of Table 3, we test whether households facing greater rural income risk—controlling for average rural income—are more likely to migrate. We find that they are: a one-standard-deviation increase in rural income risk predicts 10% more migrants ($p < 0.01$). Our measure of income risk is the within-household standard deviation of rural household income from all survey waves prior to the endline survey (that is, census, baseline, and follow-up), and our control for average rural income is the within-household average of these same three measures.

As shown in the bottom panel of Table 3, including our standard set of demographic controls leaves these results largely unchanged, though the coefficient on rural income risk is attenuated (coeff = 6% more migrants per standard deviation of risk, $p < 0.10$).

4.4 Shocking Households' Information About Nairobi

Finally, we study how households' linking and migration decisions change in the face of exogenous information shocks about potential earnings in Nairobi. To do so, we exploit the randomized controlled trial described in Barnett-Howell et al. (2026), which randomly assigned villages in our dataset to receive information about Nairobi or not. Our model predicts that new information will be a substitute for origin advice connections: we therefore interact the treatment assignment variable with the baseline (pre-experimental) number of advice connections.

Table 4 presents results. Column 1 shows differential impacts of the shock on migration

to Nairobi between $t = 1$ (the shock) and $t = 2$ (the endline, one year later). We find that the shock increases migration significantly less for well-connected households (interaction term $= -14\%$; $p = 0.03$). The impact of the uninteracted shock (on unconnected households) is positive (14%; $p = 0.07$). This is consistent with the first part of hypothesis 6: that information shocks will affect poorly connected households more than well-connected ones.

Columns 2–4 of Table 4 investigate the second part of hypothesis 6: how the information shock affects link formation in the village. We find a slight substitution away from origin information for well-connected households, but the difference is not statistically significant. Interestingly, the starkest substitution appears for financial and farm-assistance links: the impact of the uninteracted shock on the number of these links is -15% and -17% respectively (p-values < 0.10). These changes are consistent with a substitution away from non-informational relationships among relatively unconnected households exposed to the information shock. For these households, as new information decreases the value of new links, we observe a shift away from the two most common links in our data: those used for financial or farming assistance. One potential explanation for why these patterns do not hold for information links is that for some households, the information shock may have been complementary with local conversations: for example, learning general information about Nairobi may have increased the value of specific information about job leads or neighborhood quality. Because our measure of information links at time $t = 2$ measures the flow of information through actual conversations rather than the hypothetical stock of connections, it is more likely to pick up changes in the intensity of use of existing links.

Table 3: Correlates of Migration

	# Nairobi Migrants in Period 2	
No Controls		
Nairobi Income in Period 1	0.10***	
	(0.01)	
	[0.00]	
Rural Income in Period 1	-0.06**	
	(0.03)	
	[0.02]	
Rural Income Risk		0.08***
		(0.04)
		[0.01]
+ HH Demographic Controls		
Nairobi Income in Period 1	0.07***	
	(0.01)	
	[0.00]	
Rural Income in Period 1	-0.05*	
	(0.02)	
	[0.06]	
Rural Income Risk		0.06*
		(0.04)
		[0.10]
Dep. Var. Mean	0.25	0.25
Observations	16,339	15,447

All models use Poisson regression. Each observation is a household. Dependent variable is the number of migrants in Nairobi at endline, or who traveled to Nairobi temporarily between baseline and endline. *Rural Income Risk* is the within-household standard deviation of rural income measured at three points prior to the endline survey. All income units are standardized to mean 0, standard deviation 1. Regressions in column 1 control for the number of Nairobi migrants at baseline (period 1) to remove the time-invariant correlation between migration and migration income. Regressions in column 2 control for average rural income. Heteroskedasticity-robust standard errors in parentheses; two-sided p-values in brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Impacts of Information Shocks on Links and Migration

	# Nairobi Migrants	# Information Links	# Financial Links	# Farm Assistance Links
Information Shock × Initial Information Connections	-0.14**	-0.09	0.06	0.06
	(0.06)	(0.09)	(0.06)	(0.07)
	[0.03]	[0.29]	[0.30]	[0.39]
Information Shock	0.14*	0.11	-0.15*	-0.17*
	(0.08)	(0.12)	(0.08)	(0.10)
	[0.07]	[0.39]	[0.07]	[0.09]
Initial Information Connections	0.15***	0.19***	0.03	0.03
	(0.05)	(0.07)	(0.05)	(0.06)
	[0.00]	[0.01]	[0.52]	[0.62]
Dep. Var. Mean	0.26	0.16	0.32	0.25
Observations	9,913	9,739	9,748	9,751

All models use Poisson regression. Each observation is a household. Information links are social connections with whom the household discussed migrating to Nairobi. Financial (farm assistance) links are households the respondent would consider going to to ask for a loan (help with farming). Number of Nairobi migrants is measured at endline (after the information shock). *Information Shock* is the randomized assignment to receive information from Barnett-Howell et al. (2026). *Initial Information Connections* is the number of households the respondents could potentially go to for information about Nairobi as of the baseline survey. Standard errors in parentheses are clustered at the village level; two-sided p-values in brackets. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Dyadic Analysis

The previous analysis evaluated correlations between a household's characteristics and its number of social connections or interactions. However, our model also makes predictions about the characteristics of a household's social connections rather than merely about the number of those connections. To study the influence of these characteristics, we exploit survey data on the identities of specific links.[15] We construct a dyadic dataset in which each observation is a potential new pairing ij—meaning, a pair of households i and j residing in the same village but unconnected at baseline. We examine whether the pair forms a new financial connection, or any new social connection, between baseline and endline. We restrict the sample to dyads in which household j had migrated by baseline, in line with hypothesis 7.

Table 5 examines how new-link formation depends on household j's baseline connectedness. The model predicts that, holding migration exposure fixed, a more highly connected potential partner is less attractive because that partner's income is divided across a larger number of relationships. Columns 1 and 3 first estimate the reduced-form relationship between j's total degree and new-link formation. Columns 2 and 4 additionally control for the number of j's baseline peers who had migrated. This distinction is useful because migration exposure, d_{j1}^{M}, affects the precision of the migration information that j can transmit, whereas total degree, d_{j1}, governs the dilution of the income-sharing benefit.

For households not planning to migrate, j's degree is negatively associated with formation of a new financial link. The association becomes somewhat larger when we hold j's migration exposure fixed, changing from -0.02 ($p = 0.20$) to -0.03 ($p = 0.08$). This pattern is suggestive of the financial mechanism in the model: conditional on the potentially informational component of j's network, households are less likely to form financial relationships with partners whose resources are spread across more connections. The corresponding estimates for formation of any new link are also negative but smaller and statistically uncertain. One potential explanation is that financial relationships are more intensive than other social connections, so the dilution of a partner's resources across additional links may matter more for financial relationships—a margin our model of binary link formation does not capture.

We also interact j's degree with an indicator for whether household i reports that it might

[15] These data are incomplete. During surveys, enumerators asked respondents to list all the other households in their village with whom they had some particular relationship. The enumerator recorded the number of connections in all cases. The enumerator also attempted to electronically search for each connection within a database of names covering the respondent's village, taken from an earlier census survey. In cases where a match was found, the enumerator's device recorded a unique ID of that connection, allowing us to link them in our data. In cases where no match was found, the enumerator recorded the name given as a string, which we later attempted to map to household IDs using a fuzzy matching technique. If no match could be made, that link is excluded from our dataset.

migrate. The interaction is positive in all specifications, implying that the negative relationship between j's connectedness and link formation is attenuated for households for whom migration information may be more valuable. For financial links, the interaction is 0.035 without the migrant-peer control ($p = 0.12$) and 0.034 with it ($p = 0.13$); the corresponding estimates for any new link are smaller and less precise. Although stated migration intentions are an imperfect analogue of the model's migration decision, the direction of these estimates is consistent with informational considerations partially offsetting the financial disadvantage of linking to a highly connected partner.

Table 5: Baseline Connectedness and New Link Formation

	New Financial Link		Any New Link	
Baseline Connections of HH i	0.0505***	0.0504***	0.0574**	0.0573**
	(0.0176)	(0.0176)	(0.0238)	(0.0238)
	[0.00]	[0.00]	[0.02]	[0.02]
Baseline Connections of HH j	-0.0214	-0.0286*	-0.0171	-0.0223
	(0.0166)	(0.0162)	(0.0235)	(0.0242)
	[0.20]	[0.08]	[0.47]	[0.36]
HH i is Likely to Migrate	-0.0347	-0.0343	0.0680	0.0683
	(0.0433)	(0.0433)	(0.0751)	(0.0751)
	[0.42]	[0.43]	[0.37]	[0.36]
HH i is Likely to Migrate $\times$				
Baseline Connections of HH j	0.0354	0.0344	0.0320	0.0312
	(0.0225)	(0.0227)	(0.0364)	(0.0365)
	[0.12]	[0.13]	[0.38]	[0.39]
Migrated Baseline Peers of HH j		0.0671		0.0482
		(0.0617)		(0.0893)
		[0.28]		[0.59]
Dep. Var. Mean	0.400	0.400	0.888	0.888
Observations	212,176	212,176	212,176	212,176

Notes: Each observation is a potential household dyad ij. The dependent variables indicate the formation of a new financial link or any new link, respectively. Both dependent variables are binary indicators multiplied by 100; then, coefficients are expressed in percentage points. The sample is restricted to dyads where household j migrated at baseline. Baseline Connections is the sum of baseline advice and financial connections. Migrated Baseline Peers of HH j counts distinct advice or financial peers nominated by j who migrated at baseline. The interaction coefficient measures how the association between j's baseline connectedness and new-link formation differs depending on i's likelihood of migrating. All models control for household i's baseline connectedness, household incomes, and additional household characteristics. Columns (2) and (4) additionally control for the number of migrated baseline peers of household j. Village-level sampling weights are used and standard errors are clustered at the village level. Standard errors appear in parentheses and two-sided p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Conclusion and Possible Extensions

This paper studies how migration decisions and social network formation interact when households face uncertainty about migration returns and participate in informal risk-sharing arrangements. We develop a dynamic model in which social links provide information about migration opportunities but also create future income-sharing obligations. Households therefore form connections by balancing their informational and insurance value against link costs and the expected burden of redistribution.

The model generates several predictions about migration and network formation. In the informational benchmark, greater uncertainty about migration returns increases the value of acquiring information through social links when the household would otherwise migrate, while the effect is ambiguous when it would otherwise not migrate. In the full model, this informational motive interacts with the insurance and redistribution consequences of forming a relationship. Migration experience makes households more valuable as providers of information, and externally supplied information reduces the marginal value of learning through existing networks. Migration itself responds to the relative certainty equivalents of urban and rural income. Using panel and experimental data from rural Kenya, we find evidence broadly consistent with several of these predictions. Together, our findings suggest that information acquisition, migration, and informal insurance should be analyzed jointly rather than as separate decisions.

The framework also suggests two natural extensions. First, the model represents a social relationship as a single link that jointly transmits information and is part of income sharing. In the data, however, households separately report informational, financial, and other relationships such as farm assistance. Allowing these links to be chosen separately would make it possible to study whether households seek migration information from individuals whom they would prefer not to be financially connected to, and therefore to separate more directly the informational and redistribution motives for network formation. Second, the model allows new links to form at a single stage and does not allow existing relationships to be severed. With repeated observations of social relationships, a richer dynamic model could instead allow households to form, maintain, or drop links after observing migration outcomes. This would permit us to study whether migration experience changes not only the number of social connections, but also the composition of households' informational and financial networks.

More broadly, the paper highlights that social networks play dual roles in environments with imperfect information. They facilitate learning about economic opportunities while simultaneously creating redistribution obligations that influence which relationships house-

holds are willing to form. Taking both effects into account provides a richer understanding of migration behavior and shows that the effectiveness of information-based policies may depend on the pre-existing structure and the local network.

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A Appendix

This appendix collects the derivations and proofs for the theoretical results in Section 2. The order follows the main text.

A.1 Certainty Equivalents and Ex-Ante Evaluation of Information

Under CARA utility, $u(c_{it}) = -e^{-rc_{it}}$, $r > 0$, and normally distributed consumption, the certainty equivalent is $CE(c_{it}) = \mathbb{E}[c_{it}] - \frac{r}{2}\operatorname{Var}(c_{it})$. Indeed, if $c_{it} \sim \mathcal{N}(\mu_{it}, \sigma_{it}^2)$, then

$$\mathbb{E}[u(c_{it})] = -\mathbb{E}[e^{-rc_{it}}] = -\exp\left(-r\mu_{it} + \frac{r^2}{2}\sigma_{it}^2\right) = -\exp\left[-r\left(\mu_{it} - \frac{r}{2}\sigma_{it}^2\right)\right] = u(CE(c_{it}))\,.$$

The positive rural-income certainty-equivalent condition used in the main text can be stated i here:

$$CE\left(\frac{y_{it}}{d_{it}+1}\right) = \frac{\mathbb{E}[y_{it}]}{d_{it}+1} - \frac{r}{2}\frac{\operatorname{Var}(y_{it})}{(d_{it}+1)^2},$$

a sufficient condition for this expression to be positive for every $d_{it} \geq 0$ is $\mathbb{E}[y_{it}] > \frac{r}{2}\operatorname{Var}(y_{it})$. Multiplying the certainty equivalent by $d_{it}+1 > 0$ gives $\frac{\mathbb{E}[y_{it}]}{d_{it}+1} - \frac{r}{2}\frac{\operatorname{Var}(y_{it})}{d_{it}+1}$, which is bounded below by $\mathbb{E}[y_{it}] - \frac{r}{2}\operatorname{Var}(y_{it})$. Hence the condition imposed in Assumption 2 is sufficient for positivity at every feasible degree.

A distinction is important when information is observed before a subsequent action. The conditional certainty equivalent after observing a signal is an interim object. Before the signal is observed, the household evaluates the compound lottery using expected utility, not by averaging conditional certainty equivalents.

For example, suppose household i forms an information-only link with household j and observes s_{j1}^i before choosing $M_{i,2}$. The ex-ante certainty equivalent associated with the optimized continuation payoff is

$$CE_i(G + ij) = -\frac{1}{r}\log \mathbb{E}_{s_{j1}^i}\left[\exp\left\{-r \max_{M_{i,2}\in\{0,1\}} CE_i\left(M_{i,2} \mid \mathcal{I}_{i1}, s_{j1}^i\right)\right\} \,\middle|\, \mathcal{I}_{i1}\right].$$

When G contains other new links, the expectation also integrates over the signals generated by those links; these signals are left implicit to keep the notation aligned with the main text.

Now hold household i's migration decision fixed at the decision it would follow without

using the additional signal from j, denoted by $M_{i,2}^G$. Then

$$-\frac{1}{r}\log \mathbb{E}_{s_{j1}^i}\left[\exp\left\{-rCE_i\left(M_{i,2}^G \mid \mathcal{I}_{i1}, s_{j1}^i\right)\right\} \;\middle|\; \mathcal{I}_{i1}\right] = CE_i\left(M_{i,2}^G \mid \mathcal{I}_{i1}\right).$$

To see this, use

$$\exp\left\{-rCE_i\left(M_{i,2}^G \mid \mathcal{I}_{i1}, s_{j1}^i\right)\right\} = -\mathbb{E}\left[u(c_{i,2}) \mid \mathcal{I}_{i1}, s_{j1}^i, M_{i,2}^G\right]$$

and the law of iterated expectations. Therefore, if the migration action is held fixed, receiving an additional signal does not change ex-ante expected utility. In the information-only benchmark, the value of a signal comes from the possibility of conditioning the subsequent migration decision on that signal.

A.2 Posterior Updating and Proof of Corollary 1: Signals and Migration

After observing the period-1 migration outcomes available through its initial network, household i has $\mu_{i1} = \mathbb{E}[z_1 \mid \mathcal{I}_{i1}]$, $V_{i1} = \mathrm{Var}(z_1 \mid \mathcal{I}_{i1})$. For every newly linked household j that migrated in period 1, household i observes $s_{i1}^j = e_{j1} + \epsilon_{j1} = z_1 + \widetilde{e}_{j1} + \epsilon_{j1}$, where $\epsilon_{j1} \sim \mathcal{N}\left(0, \frac{\sigma_\epsilon^2}{1+d_{j1}^M}\right)$. Since $\widetilde{e}_{j1}$ and ϵ_{j1} are independent $s_{i1}^j \mid z_1 \sim \mathcal{N}\left(z_1, \sigma_{\widetilde{e}}^2 + \frac{\sigma_\epsilon^2}{1+d_{j1}^M}\right)$. Each informative new link therefore adds the precision $\frac{1}{\sigma_{\widetilde{e}}^2+\sigma_\epsilon^2/(1+d_{j1}^M)}$to household i's information about z_1. Gaussian updating gives

$$\widehat{\sigma}_{z,i,2}^2 = \left[\frac{1}{V_{i1}} + \sum_{\substack{j\in N_{i2}\setminus N_{i1}\\ M_{j1}=1}} \frac{1}{\sigma_{\widetilde{e}}^2 + \sigma_\epsilon^2/(1+d_{j1}^M)}\right]^{-1},$$

and

$$\widehat{\mu}_{z,i,2} = \widehat{\sigma}_{z,i,2}^2\left[\frac{\mu_{i1}}{V_{i1}} + \sum_{\substack{j\in N_{i2}\setminus N_{i1}\\ M_{j1}=1}} \frac{s_{i1}^j}{\sigma_{\widetilde{e}}^2 + \sigma_\epsilon^2/(1+d_{j1}^M)}\right].$$

The period-2 migration decision, conditional on the migration choices of the other house-

holds, is therefore characterized by

$$\frac{\rho\widehat{\mu}_{z,i,2}}{d_{i2}+1} - q - \frac{r}{2}\Big[\frac{\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}}{(d_{i2}+1)^2} + \frac{2\left(\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda\right)}{d_{i2}+1}\sum_{j\in N_{i2}}\frac{M_{j2}}{d_{j2}+1}\Big] > 0.$$

Since $z_2 = \rho z_1 + \lambda_2$, $e_{i2} = z_2 + \widetilde{e}_{i2}$, we obtain $\mathbb{E}[e_{i2} \mid \mathcal{I}_{i2}] = \rho\widehat{\mu}_{z,i,2}$, and $\mathrm{Var}(e_{i2} \mid \mathcal{I}_{i2}) = \rho^2\widehat{\sigma}^2_{z,i,2} + \sigma^2_\lambda + \sigma^2_{\widetilde{e}}$. Therefore household i migrates in period 2 if and only if

$$\rho\widehat{\mu}_{z,i,2} - q - \frac{r}{2}\left(\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}\right) > 0.$$

The posterior variance depends on the precision of the signal but not on its realized value. In contrast, the posterior mean is increasing in a favorable realization. For a signal from household j,

$$\frac{\partial\widehat{\mu}_{z,i,2}}{\partial s^j_{i1}} = \frac{\widehat{\sigma}^2_{z,i,2}}{\sigma^2_{\widetilde{e}}+\sigma^2_\epsilon/(1+d^M_{j1})} > 0.$$

Hence a favorable signal raises expected urban income without increasing the posterior variance of urban income. It raises the period-2 migration propensity. This proves Corollary 1.

The same expression also shows why migration exposure makes household j a more precise source of information. Since

$$\frac{\partial}{\partial d^M_{j1}}\left(\sigma^2_{\widetilde{e}} + \frac{\sigma^2_\epsilon}{1+d^M_{j1}}\right) = -\frac{\sigma^2_\epsilon}{(1+d^M_{j1})^2} < 0,$$

greater period-1 migration exposure lowers communication noise and increases the precision of the signal transmitted by j.

A.3 Proof of Proposition 1: Strategic Substitutability in Period-1 Migration

Household i chooses M_{i1} before period-1 income realizations are observed. Its migration incentive must therefore be evaluated in expected-utility units $\phi_{i1} = \Delta\mathbb{E}[u(c_{i1})]+\beta\Delta\mathbb{E}[u(c_{i2})]$, where Δ denotes the difference between $M_{i1} = 1$ and $M_{i1} = 0$. Household i migrates whenever $\phi_{i1} > 0$.

Consider a neighbor j and hold fixed the period-1 migration choices of all households other than i and j. Period-1 migration choices are strategic substitutes if and only if $\phi_{i1}|_{M_{j1}=1} \leq \phi_{i1}|_{M_{j1}=0}$. The comparison contains a contemporaneous component and a con-

tinuation component:

$$\begin{aligned}&\phi_{i1}|_{M_{j1}=1} - \phi_{i1}|_{M_{j1}=0}\\ =\ &\left[\Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=1} - \Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=0}\right] + \beta\left[\Delta\mathbb{E}[u(c_{i2})]|_{M_{j1}=1} - \Delta\mathbb{E}[u(c_{i2})]|_{M_{j1}=0}\right].\end{aligned}$$

We study the two terms separately.

Using the income-sharing rule, household i's period-1 consumption is

$$c_{i1} = \frac{y_{i1} + e_{i1}M_{i1}}{d_{i1}+1} + \sum_{k\in N_{i1}} \frac{y_{k1} + e_{k1}M_{k1}}{d_{k1}+1} - qM_{i1}.$$

Conditional on the migration profile, expected consumption is

$$\mathbb{E}[c_{i1}] = \frac{\bar{y}}{d_{i1}+1} + \sum_{k\in N_{i1}} \frac{\bar{y}}{d_{k1}+1} + \bar{z}\left[\frac{M_{i1}}{d_{i1}+1} + \sum_{k\in N_{i1}} \frac{M_{k1}}{d_{k1}+1}\right] - qM_{i1},$$

and its variance is

$$\mathrm{Var}(c_{i1}) = \frac{\sigma^2}{(d_{i1}+1)^2} + \sum_{k\in N_{i1}} \frac{\sigma^2}{(d_{k1}+1)^2} + \sigma_z^2\left[\frac{M_{i1}}{d_{i1}+1} + \sum_{k\in N_{i1}} \frac{M_{k1}}{d_{k1}+1}\right]^2 + \sigma_{\tilde{e}}^2\left[\frac{M_{i1}}{(d_{i1}+1)^2} + \sum_{k\in N_{i1}} \frac{M_{k1}}{(d_{k1}+1)^2}\right].$$

Since consumption is normal, $CE_i = \mathbb{E}[c_{i1}] - \frac{r}{2}\mathrm{Var}(c_{i1})$. The certainty-equivalent gain from household i's own migration, conditional on the migration choices of its neighbors, is therefore

$$\begin{aligned}&CE_i(M_{i1}=1) - CE_i(M_{i1}=0)\\ =\ &\frac{\bar{z}}{d_{i1}+1} - q - \frac{r}{2}\left[\sigma_z^2\left(\frac{1}{(d_{i1}+1)^2} + \frac{2}{d_{i1}+1}\sum_{k\in N_{i1}}\frac{M_{k1}}{d_{k1}+1}\right) + \frac{\sigma_{\tilde{e}}^2}{(d_{i1}+1)^2}\right].\end{aligned}$$

Now compare $M_{j1}=1$ with $M_{j1}=0$. All terms that do not contain M_{j1} cancel, giving

$$[CE_i(1 \mid M_{j1}=1) - CE_i(0 \mid M_{j1}=1)] - [CE_i(1 \mid M_{j1}=0) - CE_i(0 \mid M_{j1}=0)] = -\frac{r\sigma_z^2}{(d_{i1}+1)(d_{j1}+1)} < 0.$$

Thus household j's migration lowers the certainty-equivalent gain from household i's own migration. The reason is the common component z_1. If j migrates, household i already receives migration income exposed to z_1. If i also migrates, its own urban income introduces an additional exposure to the same common shock. The additional variance created by this

covariance is

$$\frac{2\sigma_z^2}{(d_{i1}+1)(d_{j1}+1)},$$

which lowers the certainty equivalent of household i's own migration by

$$\frac{r}{2}\frac{2\sigma_z^2}{(d_{i1}+1)(d_{j1}+1)} = \frac{r\sigma_z^2}{(d_{i1}+1)(d_{j1}+1)}.$$

Because the period-1 objective is expressed in expected-utility rather than certainty-equivalent differences, the contemporaneous condition relevant for the full intertemporal comparison is

$$\Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=1} \leq \Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=0}\,.$$

The calculation above provides the primitive covariance force generating this inequality. In particular, it is sufficient that migration by j weakly raises household i's non-migration certainty equivalent while the reduction in the certainty-equivalent gain from household i's migration derived above is not offset by a level effect through the curvature of $u(\cdot)$.

Period-1 migration also generates information that household i can use when choosing M_{i2}. The maximization over M_{i2} must therefore occur after the period-1 information has been observed.

For every information set available before the period-2 migration decision,

$$\max_{M_{i2}\in\{0,1\}} \mathbb{E}[u(c_{i2}) \mid M_{i2},\ \text{information}] = u\left(\max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid\ \text{information})\right),$$

because under CARA-normality

$$\mathbb{E}[u(c_{i2}) \mid M_{i2},\ \text{information}] = u\left(CE_i(M_{i2} \mid\ \text{information})\right),$$

and $u(\cdot)$ is strictly increasing.

Suppose first that $M_{j1}=0$. If $M_{i1}=0$, household i does not observe e_{i1}. If $M_{i1}=1$, it observes e_{i1} and can condition M_{i2} on this realization. The period-2 option value generated by household i's own period-1 migration is therefore

$$\mathcal{O}_{i,1}|_{M_{j1}=0} = \mathbb{E}_{e_{i1}}\left[u\left(\max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid e_{i1})\right)\right] - u\left(\max_{M_{i2}\in\{0,1\}} CE_i(M_{i2})\right).$$

Now suppose that $M_{j1}=1$. Household i observes e_{j1} even if it does not migrate. If it

also migrates, it additionally observes e_{i1}. Hence

$$\mathcal{O}_{i,1}|_{M_{j1}=1} = \mathbb{E}_{e_{j1}}\Big[\mathbb{E}_{e_{i1}|e_{j1}}\left[u\left(\max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid e_{j1}, e_{i1})\right)\right] - u\left(\max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid e_{j1})\right)\Big].$$

Thus, $\mathcal{O}_{i,1}|_{M_{j1}=0}$ = value of moving from no new migration observation to e_{i1}, whereas $\mathcal{O}_{i,1}|_{M_{j1}=1}$ = value of adding e_{i1} after e_{j1} is already observed. Both option values are non-negative. For every realization of the additional information, household i can always retain the period-2 migration action it would have chosen without that information. Reoptimizing after observing the additional signal cannot lower expected continuation utility.

The direct period-1 migration outcomes satisfy

$$e_{i1} = z_1 + \widetilde{e}_{i1}, \qquad e_{j1} = z_1 + \widetilde{e}_{j1}.$$

If neither outcome has yet been observed, $\mathrm{Var}(z_1) = \sigma_z^2$. After observing one migration outcome,

$$\mathrm{Var}(z_1 \mid e_{i1}) = \mathrm{Var}(z_1 \mid e_{j1}) = \left(\frac{1}{\sigma_z^2} + \frac{1}{\sigma_{\widetilde{e}}^2}\right)^{-1}.$$

After observing both,

$$\mathrm{Var}(z_1 \mid e_{j1}, e_{i1}) = \left(\frac{1}{\sigma_z^2} + \frac{2}{\sigma_{\widetilde{e}}^2}\right)^{-1}.$$

The reduction in posterior variance produced by the first observation is

$$\sigma_z^2 - \left(\frac{1}{\sigma_z^2} + \frac{1}{\sigma_{\widetilde{e}}^2}\right)^{-1} = \frac{\sigma_z^4}{\sigma_{\widetilde{e}}^2 + \sigma_z^2}.$$

The additional reduction produced by the second observation is

$$\left(\frac{1}{\sigma_z^2} + \frac{1}{\sigma_{\widetilde{e}}^2}\right)^{-1} - \left(\frac{1}{\sigma_z^2} + \frac{2}{\sigma_{\widetilde{e}}^2}\right)^{-1} = \frac{\sigma_z^4 \sigma_{\widetilde{e}}^2}{(\sigma_{\widetilde{e}}^2 + \sigma_z^2)(\sigma_{\widetilde{e}}^2 + 2\sigma_z^2)}.$$

Their ratio is $\frac{\sigma_{\widetilde{e}}^2}{\sigma_{\widetilde{e}}^2 + 2\sigma_z^2} < 1$. Therefore,

$$\mathrm{Var}(z_1 \mid e_{j1}) - \mathrm{Var}(z_1 \mid e_{j1}, e_{i1}) < \mathrm{Var}(z_1) - \mathrm{Var}(z_1 \mid e_{i1}).$$

Since $e_{i2} = \rho z_1 + \lambda_2 + \widetilde{e}_{i2}$, then the uncertainty about period-2 urban income:

$$\mathrm{Var}(e_{i2} \mid e_{j1}) - \mathrm{Var}(e_{i2} \mid e_{j1}, e_{i1}) < \mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid e_{i1}).$$

Hence household i's own period-1 migration outcome always produces a smaller additional reduction in uncertainty when household j has already migrated. This is a global statistical-substitution result.

The previous inequality does not by itself imply $\mathcal{O}_{i,1}|_{M_{j1}=1} \leq \mathcal{O}_{i,1}|_{M_{j1}=0}$, because e_{j1} changes both the posterior variance and the posterior mean. In particular, a realization of e_{j1} can move household i closer to the period-2 migration cutoff, in which case the second observation may be especially useful for deciding whether $M_{i2} = 0$ or $M_{i2} = 1$.

The decision-value comparison can nevertheless be proved under transparent parameter restrictions. Consider first the information-only continuation benchmark, so that an additional period-1 migration observation affects the period-2 problem only through beliefs about e_{i2}.

To characterize the option value more explicitly, it is useful to rewrite the period-2 optimization in terms of the difference between the certainty equivalents from not migrating and migrating. Since $M_{i2} \in \{0, 1\}$, $\max_{M_{i2}\in\{0,1\}} CE_i(M_{i2}) = CE_i(0) - \min\{CE_i(0) - CE_i(1), 0\}$. The difference $CE_i(0) - CE_i(1)$ therefore completely determines the period-2 migration decision:

$$CE_i(0) - CE_i(1) \begin{cases} > 0 & \Longrightarrow M_{i2} = 0, \\ = 0 & \Longrightarrow \text{household } i \text{ is indifferent}, \\ < 0 & \Longrightarrow M_{i2} = 1. \end{cases}$$

Hence, the option value of the additional information can equivalently be expressed in terms of how the signal changes $CE_i(0) - CE_i(1)$, that is, how it changes household i's position relative to the period-2 migration threshold.

Under the information-only benchmark, the non-migration certainty equivalent $CE_i(0)$ is unaffected by the migration signals. Using CARA utility, $u(c) = -\exp(-rc)$, it follows that

$$\begin{aligned} u\left(\max_{M_{i2}} CE_i(M_{i2})\right) &= -\exp\left\{-r\left[CE_i(0) - \min\{CE_i(0) - CE_i(1), 0\}\right]\right\} \\ &= -e^{-rCE_i(0)} \exp\left\{r \min\{CE_i(0) - CE_i(1), 0\}\right\}. \end{aligned}$$

This representation is useful because the common term $e^{-rCE_i(0)}$ can be factored out, while

the remaining term captures only whether and by how much the information changes the relative attractiveness of migration.

Before receiving the candidate migration observations, the period-2 certainty-equivalent migration gain is $CE_i(1) - CE_i(0) = \rho\bar{z} - q - \frac{r}{2}\left(\rho^2\sigma_z^2 + \sigma_\lambda^2 + \sigma_{\tilde{e}}^2\right)$.

For any collection of period-1 observations, let the amount of uncertainty resolved about e_{i2} be as $\mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information})$. Before the realizations are observed, the optimized period-2 expected utility is

$$
\begin{aligned}
-e^{-rCE_i(0)}\Bigg[\Phi\left(-\frac{CE_i(1) - CE_i(0) + \frac{r}{2}[\mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information})]}{\sqrt{\mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information})}}\right) \\
+ e^{-r[CE_i(1) - CE_i(0)]}\Phi\left(\frac{CE_i(1) - CE_i(0) - \frac{r}{2}[\mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information})]}{\sqrt{\mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information})}}\right)\Bigg].
\end{aligned}
$$

This expression is increasing in the amount of uncertainty resolved. Its second derivative is non-positive whenever

$$
[CE_i(1) - CE_i(0)]^2 \leq \mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information}) + \frac{r^2}{4}\left[\mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information})\right]^2.
$$

Hence this condition guarantees locally diminishing decision value of additional information.

A particularly transparent sufficient case is the period-2 migration cutoff,

$$
CE_i(1) - CE_i(0) = 0 \rightarrow \phi_{i,2} = 0
$$

At the cutoff the optimized continuation value simplifies to

$$
-2e^{-rCE_i(0)}\Phi\left(-\frac{r}{2}\sqrt{\mathrm{Var}(e_{i2}) - \mathrm{Var}(e_{i2} \mid \text{information})}\right)
$$

. It is strictly increasing and strictly concave in the amount of uncertainty resolved.

The first migration observation resolves $\rho^2\left[\sigma_z^2 - \left(\frac{1}{\sigma_z^2} + \frac{1}{\sigma_{\tilde{e}}^2}\right)^{-1}\right]$ of the uncertainty about e_{i2}, whereas the additional amount resolved by the second observation is

$$
\rho^2\left[\left(\frac{1}{\sigma_z^2} + \frac{1}{\sigma_{\tilde{e}}^2}\right)^{-1} - \left(\frac{1}{\sigma_z^2} + \frac{2}{\sigma_{\tilde{e}}^2}\right)^{-1}\right],
$$

and the latter is strictly smaller than the former.

Because the continuation value is increasing and strictly concave at the cutoff, and because the second observation resolves strictly less additional uncertainty than the first, $\mathcal{O}_{i,1}|_{M_{j1}=1} < \mathcal{O}_{i,1}|_{M_{j1}=0}$. Thus, at the period-2 migration cutoff, household j's migration

strictly reduces the informational continuation value of household i's own period-1 migration.

The inequality is not restricted to exact indifference. Both option values are continuous in $CE_i(1) - CE_i(0)$. Since the inequality is strict when $CE_i(1) - CE_i(0) = 0$, there exists a neighborhood of the period-2 cutoff in which $\mathcal{O}_{i,1}|_{M_{j1}=1} < \mathcal{O}_{i,1}|_{M_{j1}=0}$ continues to hold.

Away from this region the decision-value comparison is not globally signed. A first signal from j may move household i sufficiently close to the period-2 migration threshold that the remaining information in e_{i1} becomes especially useful, even though it produces a smaller additional reduction in posterior variance. Gaussian learning therefore establishes diminishing statistical information globally, but diminishing optimized decision value requires the curvature condition above or another sufficient restriction.

A.3.1 Combining the two channels

For the informational continuation channel, $\Delta\mathbb{E}[u(c_{i2})]|_{M_{j1}=0} = \mathcal{O}_{i,1}|_{M_{j1}=0}$, and $\Delta\mathbb{E}[u(c_{i2})]|_{M_{j1}=1} = \mathcal{O}_{i,1}|_{M_{j1}=1}$, when the remaining continuation environment is held fixed.

Therefore, under the contemporaneous expected-utility condition

$$\Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=1} \leq \Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=0}\,,$$

and the continuation condition $\mathcal{O}_{i,1}|_{M_{j1}=1} \leq \mathcal{O}_{i,1}|_{M_{j1}=0}$, we obtain

$$\begin{aligned} &\phi_{i1}|_{M_{j1}=1} - \phi_{i1}|_{M_{j1}=0} \\ =\ &\Big[\Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=1} - \Delta\mathbb{E}[u(c_{i1})]|_{M_{j1}=0}\Big] + \beta\Big[\mathcal{O}_{i,1}|_{M_{j1}=1} - \mathcal{O}_{i,1}|_{M_{j1}=0}\Big] \leq 0. \end{aligned}$$

Hence household i's incentive to migrate in period 1 weakly decreases when household j migrates. Period-1 migration choices are therefore strategic substitutes under the stated conditions.

The statistical part of the continuation result is unconditional: another migrating neighbor always makes household i's own migration outcome less informative at the margin. The stronger decision-value result is proved, for example, at and in a neighborhood of the period-2 migration cutoff in the information-only continuation benchmark. In the full risk-sharing continuation problem, peer migration income and subsequent network formation can also depend on the realized information. In that case the corresponding option-value inequality above is the sufficient condition required for the result.

Applying the same pairwise comparison as additional neighbors migrate gives a monotone best response to neighbors' migration choices. If the relevant neighbors are symmetric in

their sharing weights and informational precision, household i's migration incentive depends monotonically on the number of migrating neighbors, yielding a household-specific cutoff $\bar{M}_i$. With heterogeneous neighbors, the pairwise strategic-substitution result remains valid, while the natural cutoff is in migration exposure rather than necessarily in the unweighted number of migrating neighbors.

A.4 Proof of Proposition 2

Starting from $\mathbb{E}[u(c)] = u(CE(c))$, and $u(.)$ is increasing. A higher certainty equivalent of income raises expected utility. Then, for changes in urban income, the result is direct. Holding the information structure and all other components of the payoff fixed, an increase in the certainty equivalent of migration income raises the expected utility of $M_{i1} = 1$ relative to $M_{i1} = 0$. Hence, under the conditions stated in the proposition, $\frac{\partial \phi_{i,1}}{\partial CE(e_{i1})} > 0$. In particular, holding expected urban income fixed, greater urban-income variance lowers $CE(e_{i1})$ and therefore lowers the migration incentive.

For rural income y_{it}, the sign requires the condition stated in the main text that is a positive CE of rural income in Assumption 2. Take the utility at period $t = 1$ and derive

$$\frac{\partial \phi_{i,1}}{\partial CE(y_{i1})} < 0 \iff (\mathbb{E}\{e^{-r(e_{it}M_{it}-q)}\} - 1) < 0 \quad \forall t$$

Economically, the condition says that a safer or more attractive rural outside option discourages migration. We can then check for the rural uncertainty in period 1 For all $j \in \bar{N}_{it}$,

$$\frac{\partial \phi_{it}}{\partial \operatorname{Var}(y_{jt})} > 0 \quad \iff \quad (\mathbb{E}\{e^{-r(e_{it}M_{it}-q)}\} - 1) < 0\,.$$

A.5 Proof of Corollary 2

Household i's period-1 migration decision must account both for the current expected-utility consequences of migration and for its effect on the continuation problem in period 2. In particular, period-1 migration may generate information that is subsequently useful when household i chooses whether to migrate in period 2.

The period-1 migration incentive is

$$\begin{aligned}\phi_{i1} = {} & \mathbb{E}\left[u(c_{i1}) \mid M_{i1} = 1\right] - \mathbb{E}\left[u(c_{i1}) \mid M_{i1} = 0\right] \\ & + \beta\Big\{\mathbb{E}\left[\max_{M_{i2}\in\{0,1\}} \mathbb{E}\left[u(c_{i2}) \mid \mathcal{I}_{i2}, M_{i1} = 1\right] \mid \mathcal{I}_{i1}\right] - \mathbb{E}\left[\max_{M_{i2}\in\{0,1\}} \mathbb{E}\left[u(c_{i2}) \mid \mathcal{I}_{i2}, M_{i1} = 0\right] b \mid \mathcal{I}_{i1}\right]\Big\}.\end{aligned} \tag{2}$$

The continuation terms in (2) already incorporate the value of information generated by period-1 migration. In particular, after the information available in period 2 is realized, household i is free to choose the period-2 migration decision that maximizes its conditional expected utility. Thus no additional option-value term should be added separately to (2).

Consider first the contemporaneous component. A sufficient condition for isolation to increase the current expected-utility gain from migration is

$$\begin{aligned}&\mathbb{E}\left[u(c_{i1}) \mid M_{i1}=1,\ N_{i1}=\emptyset\right]-\mathbb{E}\left[u(c_{i1}) \mid M_{i1}=0,\ N_{i1}=\emptyset\right]\\&\quad\geq \mathbb{E}\left[u(c_{i1}) \mid M_{i1}=1,\ N_{i1}\neq\emptyset\right]-\mathbb{E}\left[u(c_{i1}) \mid M_{i1}=0,\ N_{i1}\neq\emptyset\right].\end{aligned} \tag{3}$$

The continuation component additionally contains the informational value of household i's own period-1 migration. If household i is isolated, it cannot obtain period-1 migration information from pre-existing neighbors. By contrast, when $N_{i1}\neq\emptyset$, migration by neighboring households may already provide information about migration returns.

Hence, under the condition that the marginal decision value of household i's own migration experiment is weakly decreasing in the information already available to the household:

$$\begin{aligned}&\mathbb{E}\left[\max_{M_{i2}}\mathbb{E}\left[u(c_{i2}) \mid \mathcal{I}_{i2}, M_{i1}=1, N_{i1}=\emptyset\right] \mid \mathcal{I}_{i1}\right]-\mathbb{E}\left[\max_{M_{i2}}\mathbb{E}\left[u(c_{i2}) \mid \mathcal{I}_{i2}, M_{i1}=0, N_{i1}=\emptyset\right] \mid \mathcal{I}_{i1}\right]\\&\quad\geq \mathbb{E}\left[\max_{M_{i2}}\mathbb{E}\left[u(c_{i2}) \mid \mathcal{I}_{i2}, M_{i1}=1, N_{i1}\neq\emptyset\right] \mid \mathcal{I}_{i1}\right]-\mathbb{E}\left[\max_{M_{i2}}\mathbb{E}\left[u(c_{i2}) \mid \mathcal{I}_{i2}, M_{i1}=0, N_{i1}\neq\emptyset\right] \mid \mathcal{I}_{i1}\right].\end{aligned} \tag{4}$$

The intuition is that period-1 migration is partly an experiment. An isolated household has fewer alternative sources of information and therefore has more to learn from its own migration experience. However, a connected household can learn from the migration experiences of its neighbors, reducing the marginal continuation value of experimenting through its own migration.

Gaussian learning supports the first part of this argument: for a signal of given precision, the reduction in posterior uncertainty is larger when the household begins with less precise information. The corresponding inequality in optimized continuation utility additionally requires the diminishing-decision-value condition in (4), because a larger reduction in posterior variance does not by itself guarantee a larger value of information for a discrete migration decision.

Combining (3) and (4) gives $\phi_{i1}|_{N_{i1}=\emptyset}\geq\phi_{i1}|_{N_{i1}\neq\emptyset}$. The inequality is strict whenever either the contemporaneous inequality or the continuation-value inequality is strict. Therefore, under the stated conditions, isolation weakly increases household i's incentive to migrate in

period 1. This proves Corollary 2.

A.6 Option Value of an Additional Link

The option value has the same definition in the information-only benchmark and in the full model. It measures the gain from allowing household i to choose its own period-2 migration action after observing the additional signal from household j, relative to retaining the migration action $M_{i,2}^G$ that would be optimal without using that additional signal. Both alternatives are evaluated under the post-link environment $G + ij$ so that the option value isolates the value of reoptimizing the migration decision rather than the direct financial consequence of the new relationship.

Thus,

$$\begin{aligned}\mathcal{O}_i(ij \mid G; r) = & -\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r \max_{M_{i,2}\in\{0,1\}} CE_i\big(M_{i,2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j\big)\right\} \mid \mathcal{I}_{i1}, G\right] \\ & + \frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r\, CE_i\big(M_{i,2}^G \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j\big)\right\} \mid \mathcal{I}_{i1}, G\right].\end{aligned}$$

When other households' period-2 migration choices affect household i's consumption, the same expression is understood conditional on those choices; at the link-formation stage the ex-ante value integrates over the migration profiles induced by equilibrium continuation strategies.

For every realization of s_{i1}^j, the baseline action $M_{i,2}^G$ remains feasible. Therefore,

$$\max_{M_{i,2}\in\{0,1\}} CE_i\big(M_{i,2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j\big) \geq CE_i\big(M_{i,2}^G \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j\big).$$

Since $r > 0$, the function $x \mapsto e^{-rx}$ is decreasing, so

$$\exp\left\{-r \max_{M_{i,2}\in\{0,1\}} CE_i\big(M_{i,2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j\big)\right\} \leq \exp\left\{-r\, CE_i\big(M_{i,2}^G \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j\big)\right\}.$$

Taking expectations preserves this inequality. The logarithm is increasing, while multiplication by $-1/r$ reverses it. Therefore, the first ex-ante certainty equivalent is at least as large as the second, and $\mathcal{O}_i(ij \mid G; r) \geq 0$. The inequality is strict if after optimizing $M_{i,2}$ gives a strictly higher conditional certainty equivalent for a set of signal realizations with positive probability. If the additional signal never changes the optimal migration action, then $\mathcal{O}_i(ij \mid G; r) = 0$.[16]

[16]Notice that with an informative, nondegenerate Gaussian signal having unbounded support as the normal random variable, the signal has a positive probability of changing a finite migration cutoff. In that setting,

A.7 Proof of Proposition 3: Information-Only Link Formation

In the information-only benchmark, the new link changes household i's information but not its income-sharing arrangement. Holding $M_{i,2}$ fixed, the law of iterated expectations established above implies that the signal does not change ex-ante expected utility. Hence the gross informational value of the link is exactly the option value $\omega_{ij} = \mathcal{O}_i(ij \mid G; r) \geq 0$.

A posterior-variance reduction must not be added as a separate ex-ante welfare term. Conditional variance falls after the signal is observed, but before the signal is observed this is exactly offset, for a fixed action, by dispersion in the posterior mean across possible signal realizations. Pure information creates ex-ante value only through the possibility of changing the subsequent migration action.

Under the tie-breaking convention in the main text, household i weakly agrees to form the link if and only if $\omega_{ij} \geq \kappa$, and household j weakly agrees if and only if $\omega_{ji} \geq \kappa$. The link forms when both weakly benefit and at least one strictly benefits:

$$\omega_{ij} \geq \kappa, \qquad \omega_{ji} \geq \kappa, \qquad \max\{\omega_{ij} - \kappa, \omega_{ji} - \kappa\} > 0.$$

This proves Proposition 3.

A.8 Closed-Form Information-Only Option Value

In the information-only benchmark, rural income is received under either migration action and does not depend on the signal. Conditional on the additional signal, the certainty-equivalent gain from migration relative to non-migration is $\mathbb{E}\left[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j\right] - q - \frac{r}{2} V_i(j)$. Under Gaussian updating, $V_i(j)$ does not depend on the realization of s_{i1}^j. Before the signal is observed, the posterior mean satisfies

$$\mathbb{E}_{s_{i1}^j}\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j] \mid \mathcal{I}_{i1}, G\right] = \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G],$$

and the law of total variance gives

$$V_i = \mathbb{E}_{s_{i1}^j}\left[\mathrm{Var}(e_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j) \mid \mathcal{I}_{i1}, G\right] + \mathrm{Var}_{s_{i1}^j}\left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j] \mid \mathcal{I}_{i1}, G\right).$$

Because the posterior variance is constant across signal realizations,

$$\mathrm{Var}_{s_{i1}^j}\left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j] \mid \mathcal{I}_{i1}, G\right) = V_i - V_i(j).$$

the option value is typically strictly positive, although it can be arbitrarily small. Having a value of zero is related to an uninformative or redundant signal, and then an action that cdoes not change.

Therefore the conditional migration gain is normally distributed before the signal is observed, with mean $\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)$, and variance $V_i - V_i(j)$. Household i migrates only when the conditional migration gain is positive. Hence the expectation entering the ex-ante inverse-CARA certainty equivalent is the sum of two regions: the probability of a non-positive migration gain, where the optimized gain is zero, and the CARA-weighted positive region, where the household migrates. Standardizing the normal distribution gives the first element $\Phi\left(-\frac{\mathbb{E}[e_{i2}|\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}\right)$. For the positive region, we take the integral:

$$\int_0^\infty e^{-rx}\frac{1}{\sqrt{2\pi[V_i - V_i(j)]}}\exp\left\{-\frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right)\right]^2}{2[V_i - V_i(j)]}\right\}dx.$$

First, we can split the expectation according to whether the conditional certainty-equivalent gain from migration is non-positive or positive. Since $\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j] - q - \frac{r}{2}V_i(j)$ is normally distributed, conditional on $(\mathcal{I}_{i1}, G)$, with mean $\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)$ and variance $V_i - V_i(j)$, we can write

$$\begin{aligned}
&\mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r\max\left\{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j] - q - \frac{r}{2}V_i(j), 0\right\}\right\} \mid \mathcal{I}_{i1}, G\right] \\
&= \int_{-\infty}^0 \frac{1}{\sqrt{2\pi[V_i - V_i(j)]}}\exp\left\{-\frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right)\right]^2}{2[V_i - V_i(j)]}\right\}dx \\
&\quad + \int_0^\infty \exp(-rx)\frac{1}{\sqrt{2\pi[V_i - V_i(j)]}}\exp\left\{-\frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right)\right]^2}{2[V_i - V_i(j)]}\right\}dx.
\end{aligned}$$

The first integral corresponds to realizations for which the conditional migration gain is non-positive. In this region the maximum equals zero, so the exponential term equals one. Standardizing the normal distribution gives

$$\begin{aligned}
&\int_{-\infty}^0 \frac{1}{\sqrt{2\pi[V_i - V_i(j)]}}\exp\left\{-\frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right)\right]^2}{2[V_i - V_i(j)]}\right\}dx \\
&= \Phi\left(-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right).
\end{aligned}$$

For the second integral, completing the square gives[17]

$$-rx - \frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right)\right]^2}{2[V_i - V_i(j)]} =$$

$$= -\frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j) - r[V_i - V_i(j)]\right)\right]^2}{2[V_i - V_i(j)]} \underbrace{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right] + \frac{r^2}{2}[V_i - V_i(j)]}_{\text{correction}}.$$

The last two terms do not depend on x and can therefore be taken outside the integral. Hence the second integral becomes

$$\exp\left\{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right] + \frac{r^2}{2}[V_i - V_i(j)]\right\}$$
$$\times \int_0^\infty \frac{1}{\sqrt{2\pi[V_i - V_i(j)]}} \exp\left\{-\frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j) - r[V_i - V_i(j)]\right)\right]^2}{2[V_i - V_i(j)]}\right\} dx.$$

The remaining integral corresponds to the upper-tail probability of the normal, with variance $V_i - V_i(j)$ and mean $\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j) - r[V_i - V_i(j)]$. Therefore,

$$\int_0^\infty \frac{1}{\sqrt{2\pi[V_i - V_i(j)]}} \exp\left\{-\frac{\left[x - \left(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j) - r[V_i - V_i(j)]\right)\right]^2}{2[V_i - V_i(j)]}\right\} dx$$
$$= 1 - \Phi\left(\frac{-\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j) - r[V_i - V_i(j)]\right]}{\sqrt{V_i - V_i(j)}}\right)$$
$$= 1 - \Phi\left(-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} + r\sqrt{V_i - V_i(j)}\right).$$

Using $1 - \Phi(-x) = \Phi(x)$, this is $\Phi\left(\frac{\mathbb{E}[e_{i2}|\mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} - r\sqrt{V_i - V_i(j)}\right)$. Combining the

[17] Completing the square allows us to rewrite the product of the exponential CARA term and the normal density as a shifted normal density multiplied by a constant, and then, the remaining integral can be evaluated using the standard normal cumulative distribution function $\Phi(\cdot)$. .

two regions, this gives

$$\begin{aligned}
&\mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r\max\left\{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G+ij,s_{i1}^j]-q-\frac{r}{2}V_i(j),0\right\}\right\}\mid\mathcal{I}_{i1},G\right]\\
&=\Phi\left(-\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}\right)\\
&\quad+\exp\left\{-r\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)\right]+\frac{r^2}{2}[V_i-V_i(j)]\right\}\\
&\quad\times\Phi\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}-r\sqrt{V_i-V_i(j)}\right).
\end{aligned}$$

Without the additional signal, household i chooses $M_{i,2}^G$. Its certainty-equivalent migration gain relative to non-migration is $M_{i,2}^G\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i\right]$. Therefore the information-only option value is

$$\begin{aligned}
\mathcal{O}_i(ij\mid G;r)=&-\frac{1}{r}\log\left[\Phi\left(-\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}\right)\right.\\
&\quad+\exp\left\{-r\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)\right]+\frac{r^2}{2}[V_i-V_i(j)]\right\}\\
&\quad\left.\times\Phi\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}-r\sqrt{V_i-V_i(j)}\right)\right]-M_{i,2}^G\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i\right].
\end{aligned}\tag{5}$$

If $V_i=V_i(j)$, the signal is uninformative and the expression with the denominator $\sqrt{V_i-V_i(j)}$ should not be used. In that case, the posterior distribution and the optimal migration decision are unchanged, so directly from the definition $\mathcal{O}_i(ij\mid G;r)=0$.

The two possible baseline actions give a useful interpretation. If $M_{i,2}^G=0$, the second term in (5) is zero, and the option is valuable only for sufficiently favorable signals that induce migration. If $M_{i,2}^G=1$, the baseline migration gain is subtracted, and the option is valuable for unfavorable signals enough can make household i to avoid migration. In either case, the option value is zero if the signal never changes the migration choice.

A.9 Corollary 3: Prior Uncertainty and Signal Precision

A.9.1 Signal precision

Hold household i's prior mean and prior variance V_i fixed and vary only the precision of the additional signal. Greater precision lowers $V_i(j)$. Differentiating (5) with respect to $V_i(j)$

gives

$$\frac{\partial \mathcal{O}_i(ij \mid G;r)}{\partial V_i(j)} = -\frac{\Phi'\left(\dfrac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right)}{2\sqrt{V_i - V_i(j)}} \times \Big[\Phi\left(-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right) + \exp\Big\{-r\Big[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\Big] + \frac{r^2}{2}[V_i - V_i(j)]\Big\} \times \Phi\left(\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} - r\sqrt{V_i - V_i(j)}\right)\Big]^{-1} < 0. \tag{6}$$

The derivative is strictly negative whenever the signal is informative. Hence ihgher greater signal precision decreases $V_i(j)$ and increases $\mathcal{O}_i(ij \mid G;r)$. Since $\omega_{ij} = \mathcal{O}_i(ij \mid G;r)$ in the information-only benchmark, greater signal precision raises the gross informational value of a link.

The signal delivers the same result in primitives: $\frac{\partial}{\partial d_{j1}^M}\left(\sigma_{\tilde{e}}^2 + \frac{\sigma_\epsilon^2}{1+d_{j1}^M}\right) = -\frac{\sigma_\epsilon^2}{(1+d_{j1}^M)^2} < 0$. Conditional on household j having migration experience to transmit, greater migration exposure among j's period-1 neighbors therefore makes the signal more precise and raises the value of linking to j.

A.9.2 Prior uncertainty

Now hold the prior mean and the signal technology fixed and increase household i's uncertainty about the learnable common component. In the one-signal comparison, $V_i = \rho^2 V_{i1} + \sigma_\lambda^2 + \sigma_{\tilde{e}}^2$, and

$$V_i(j) = \rho^2\left[\frac{1}{V_{i1}} + \frac{1}{\sigma_{\tilde{e}}^2 + \sigma_\epsilon^2/(1+d_{j1}^M)}\right]^{-1} + \sigma_\lambda^2 + \sigma_{\tilde{e}}^2.$$

Differentiating the posterior variance gives

$$\frac{\partial V_i(j)}{\partial V_i} = \left[\frac{\sigma_{\tilde{e}}^2 + \sigma_\epsilon^2/(1+d_{j1}^M)}{V_{i1} + \sigma_{\tilde{e}}^2 + \sigma_\epsilon^2/(1+d_{j1}^M)}\right]^2 \in (0,1).$$

Thus an increase in prior uncertainty raises both pre-signal and post-signal variance, but it raises the pre-signal variance more. Equivalently, the dispersion of possible posterior means, $V_i - V_i(j)$ increases.

Differentiating the option value with respect to V_i while allowing $V_i(j)$ to respond gives

$$\begin{aligned}\frac{\partial \mathcal{O}_i(ij \mid G;r)}{\partial V_i} = {} & \frac{\Phi'\left(\dfrac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right)}{2\sqrt{V_i - V_i(j)}}\left(1 - \frac{\partial V_i(j)}{\partial V_i}\right) \\ & \times \left\{\mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r\max\left\{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j] - q - \frac{r}{2}V_i(j), 0\right\}\right\} \,\middle|\, \mathcal{I}_{i1}, G\right]\right\}^{-1} \\ & - \frac{r}{2}\exp\left\{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right] + \frac{r^2}{2}[V_i - V_i(j)]\right\} \\ & \times \Phi\left(\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} - r\sqrt{V_i - V_i(j)}\right) \\ & \times \left\{\mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r\max\left\{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j] - q - \frac{r}{2}V_i(j), 0\right\}\right\} \,\middle|\, \mathcal{I}_{i1}, G\right]\right\}^{-1} + \frac{r}{2}M_{i,2}^G. \end{aligned} \tag{7}$$

The expectation that appears twice in (7) is exactly the positive expectation evaluated in closed form immediately before (5). The first term is positive because $1 - \partial V_i(j)/\partial V_i > 0$. The remaining terms depend on the baseline action.

If $M_{i,2}^G = 1$, the final two terms can be combined because the expectation in (7) equals the sum of the non-migration and migration regions. The derivative becomes

$$\begin{aligned}\frac{\partial \mathcal{O}_i(ij \mid G;r)}{\partial V_i} = {} & \frac{\Phi'\left(\dfrac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right)}{2\sqrt{V_i - V_i(j)}}\left(1 - \frac{\partial V_i(j)}{\partial V_i}\right) \\ & \times \left\{\mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r\max\left\{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j] - q - \frac{r}{2}V_i(j), 0\right\}\right\} \,\middle|\, \mathcal{I}_{i1}, G\right]\right\}^{-1} \\ & + \frac{r}{2}\Phi\left(-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right) \\ & \times \left\{\mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r\max\left\{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j] - q - \frac{r}{2}V_i(j), 0\right\}\right\} \,\middle|\, \mathcal{I}_{i1}, G\right]\right\}^{-1} > 0. \end{aligned}$$

Thus, holding the prior mean and signal technology fixed, greater prior uncertainty raises the information-only link value in the region where the baseline action is migration.

If $M_{i,2}^G = 0$, the final $r/2$ term in (7) is absent. The positive increase in the dispersion of possible posterior means competes with the fact that greater prior uncertainty also makes migration itself more risky. Hence the sign is generally ambiguous. Therefore the prior-uncertainty comparative static must be stated with this qualification: at a fixed prior mean, it is positive when $M_{i,2}^G = 1$ and not generally signed when $M_{i,2}^G = 0$.

In the information-only benchmark, $\omega_{ij} = \mathcal{O}_i(ij \mid G;r)$. We can therefore study directly

how the value of the link changes with the amount of uncertainty resolved by the additional signal, $V_i - V_i(j)$. To isolate this effect, hold prior uncertainty V_i fixed. An increase in $V_i - V_i(j)$ is then equivalent to a reduction in posterior uncertainty $V_i(j)$. Hence,

$$\frac{\partial \omega_{ij}}{\partial \left[V_i - V_i(j)\right]} = -\frac{\partial \mathcal{O}_i(ij \mid G; r)}{\partial V_i(j)}. \tag{8}$$

From the derivative of the option value with respect to posterior uncertainty derived above,

$$\frac{\partial \mathcal{O}_i(ij \mid G; r)}{\partial V_i(j)} = -\frac{\Phi'\left(\frac{\mathbb{E}[e_{i2}|\mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right)}{2\sqrt{V_i - V_i(j)}} \times \Big[\Phi\left(-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right)$$
$$+ \exp\left\{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i\right]\right\} \times \Phi\left(\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} - r\sqrt{V_i - V_i(j)}\right)\Big]^{-1} < 0. \tag{9}$$

Combining (8) and (9) gives

$$\frac{\partial \omega_{ij}}{\partial \left[V_i - V_i(j)\right]} = \frac{\Phi'\left(\frac{\mathbb{E}[e_{i2}|\mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right)}{2\sqrt{V_i - V_i(j)}} \times \Big[\Phi\left(-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right) +$$
$$+ \exp\left\{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i\right]\right\} \times \Phi\left(\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} - r\sqrt{V_i - V_i(j)}\right)\Big]^{-1} > 0. \tag{10}$$

Therefore, we have that $\frac{\partial \omega_{ij}}{\partial [V_i - V_i(j)]} > 0.$, and the interpretation is direct. Holding the household's prior uncertainty V_i fixed, a larger difference $V_i - V_i(j)$ means that the new link resolves more uncertainty about period-2 migration income. The signal therefore has greater potential to change household i's optimal migration decision, which raises the option value of the link.

This comparative static should be distinguished from an increase in prior uncertainty V_i itself. When V_i changes, posterior uncertainty $V_i(j)$ also changes, so the effect is not equivalent to simply increasing $V_i - V_i(j)$ while holding V_i fixed.

We now derive the effect of r on (5). Hold $\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G], V_i, V_i(j)$ fixed and work locally in a region in which $M_{i,2}^G$ does not change.

First step is to simplify the exponential term: The exponent in the second term inside

the logarithm is

$$-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right] + \frac{r^2}{2}[V_i - V_i(j)] = -r[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q] + \frac{r^2}{2}V_i(j) + \frac{r^2}{2}V_i - \frac{r^2}{2}V_i(j)$$
$$= -r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i\right].$$

Therefore,

$$\frac{\partial}{\partial r}\exp\left\{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i\right]\right\} = \exp\left\{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i\right]\right\}\left[-(\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q) + rV_i\right].$$

Now we differentiate the first normal CDF. Since V_i and $V_i(j)$ are held fixed,

$$\frac{\partial}{\partial r}\left[-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right] = \frac{V_i(j)}{2\sqrt{V_i - V_i(j)}}.$$

Hence, using symmetry of the standard-normal density,

$$\frac{\partial}{\partial r}\Phi\left(-\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right) = \Phi'\left(\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right)\frac{V_i(j)}{2\sqrt{V_i - V_i(j)}}.$$

Now we compute the derivative of $\frac{\mathbb{E}[e_{i2}|\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}} - r\sqrt{V_i - V_i(j)}$ is $-\frac{V_i(j)}{2\sqrt{V_i-V_i(j)}} - \sqrt{V_i - V_i(j)}$.

The following identity allows the two density terms to simplify:

$$\exp\left\{-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right] + \frac{r^2}{2}[V_i - V_i(j)]\right\}$$
$$\times \Phi'\left(\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} - r\sqrt{V_i - V_i(j)}\right) = \Phi'\left(\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}}\right).$$

To verify it, use $\Phi'(x) = (2\pi)^{-1/2}e^{-x^2/2}$. The exponent on the left-hand side is

$$-r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right] + \frac{r^2}{2}[V_i - V_i(j)] - \frac{1}{2}\left[\frac{\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)}{\sqrt{V_i - V_i(j)}} - r\sqrt{V_i - V_i(j)}\right]^2.$$

Expanding the square gives

$$-\frac{1}{2}\frac{\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right]^2}{V_i - V_i(j)} + r\left[\mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G] - q - \frac{r}{2}V_i(j)\right] - \frac{r^2}{2}[V_i - V_i(j)].$$

The last two terms cancel the first two terms outside the square. The remaining exponent is

exactly $-\frac{1}{2}\frac{\left[\mathbb{E}[e_{i2}|\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)\right]^2}{V_i-V_i(j)}$, which is the exponent of the density on the right-hand side.

Using the product rule and the identity above, the derivative of the expression inside the logarithm in (5) is

$$\exp\left\{-r\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i\right]\right\}\left[-(\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q)+rV_i\right]$$
$$\times\,\Phi\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}-r\sqrt{V_i-V_i(j)}\right)-\sqrt{V_i-V_i(j)}\Phi'\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}\right).$$

The terms proportional to $V_i(j)/(2\sqrt{V_i-V_i(j)})$ cancel exactly.

For any positive expression inside the logarithm, $\frac{\partial}{\partial r}\left[-\frac{1}{r}\log(\cdot)\right]=\frac{1}{r^2}\log(\cdot)-\frac{1}{r}\frac{\partial_r(\cdot)}{(\cdot)}$. Moreover, locally holding $M^G_{i,2}$ fixed,

$$\frac{\partial}{\partial r}\left\{-M^G_{i,2}\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i\right]\right\}=\frac{1}{2}M^G_{i,2}V_i.$$

Combining all terms gives

$$\begin{aligned}\frac{\partial\mathcal{O}_i(ij\mid G;r)}{\partial r}=\frac{1}{r^2}\log\Bigg[&\Phi\left(-\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}\right)\\&+\exp\left\{-r\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i\right]\right\}\Phi\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}-r\sqrt{V_i-V_i(j)}\right)\Bigg]\\&-\frac{1}{r}\Bigg\{\exp\left\{-r\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i\right]\right\}\left[-\left(\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q\right)+rV_i\right]\\&\times\Phi\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}-r\sqrt{V_i-V_i(j)}\right)\\&-\sqrt{V_i-V_i(j)}\Phi'\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}\right)\Bigg\}\\&\times\Bigg[\Phi\left(-\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}\right)\\&+\exp\left\{-r\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i\right]\right\}\Phi\left(\frac{\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G]-q-\frac{r}{2}V_i(j)}{\sqrt{V_i-V_i(j)}}-r\sqrt{V_i-V_i(j)}\right)\Bigg]^{-1}\\&+\frac{1}{2}M^G_{i,2}V_i.\end{aligned}\tag{11}$$

For every realization of the signal, greater absolute risk aversion lowers the conditional certainty-equivalent gain from migration:

$$\frac{\partial}{\partial r}\left[\mathbb{E}[e_{i2}\mid\mathcal{I}_{i1},G+ij,s^j_{i1}]-q-\frac{r}{2}V_i(j)\right]=-\frac{1}{2}V_i(j)<0.$$

Thus greater risk aversion shifts the signal-contingent migration decision against migration. This effect alone does not sign (11), however, because r also changes the inverse-CARA valuation across signal realizations and, when $M^G_{i,2} = 1$, the value of the baseline migration action.

The derivative in (11) is interpreted locally within a region in which $M^G_{i,2}$ remains unchanged. At a value of r at which the baseline migration choice changes, the option-value function may have a kink.

When $M^G_{i,2} = 0$, the baseline-action term vanishes and the option value reduces to

$$\mathcal{O}_i(ij \mid G; r) = -\frac{1}{r} \log \mathbb{E}_{s^j_{i1}} \Big[\exp \Big\{ -r \max \Big\{ \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s^j_{i1}] - q - \frac{r}{2} V_i(j), 0 \Big\} \Big\} \Big| \mathcal{I}_{i1}, G \Big].$$

Consider two levels of absolute risk aversion $r' > r > 0$. There are two effects of increasing risk aversion. Now first, for every realization of s^j_{i1},

$$\max \Big\{ \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s^j_{i1}] - q - \frac{r'}{2} V_i(j), 0 \Big\} \leq \max \Big\{ \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s^j_{i1}] - q - \frac{r}{2} V_i(j), 0 \Big\}.$$

Thus, holding the inverse-CARA evaluation at r', the optimized signal-contingent payoff is weakly lower under r'.

Second, even holding the signal-contingent payoff fixed at its value under r, a larger coefficient of absolute risk aversion weakly lowers its inverse-CARA certainty equivalent. To see this, since $r' > r > 0$, $0 < \frac{r}{r'} < 1$, so $x^{r/r'}$ is concave for $x > 0$. For every realization of the signal,

$$\begin{aligned} &\exp \Big\{ -r \max \Big\{ \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s^j_{i1}] - q - \frac{r}{2} V_i(j), 0 \Big\} \Big\} \\ &= \Big[\exp \Big\{ -r' \max \Big\{ \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s^j_{i1}] - q - \frac{r}{2} V_i(j), 0 \Big\} \Big\} \Big]^{r/r'}. \end{aligned}$$

By Jensen's inequality,

$$\begin{aligned} &\mathbb{E}_{s^j_{i1}} \Big[\exp \Big\{ -r \max \Big\{ \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s^j_{i1}] - q - \frac{r}{2} V_i(j), 0 \Big\} \Big\} \Big| \mathcal{I}_{i1}, G \Big] \\ &\leq \Big\{ \mathbb{E}_{s^j_{i1}} \Big[\exp \Big\{ -r' \max \Big\{ \mathbb{E}[e_{i2} \mid \mathcal{I}_{i1}, G + ij, s^j_{i1}] - q - \frac{r}{2} V_i(j), 0 \Big\} \Big\} \Big| \mathcal{I}_{i1}, G \Big] \Big\}^{r/r'}. \end{aligned}$$

Taking logarithms and multiplying by $-1/r$ therefore gives

$$
\begin{aligned}
&-\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-r\max\Big\{\mathbb{E}[e_{i2}\mid \mathcal{I}_{i1}, G+ij, s_{i1}^j]-q-\frac{r}{2}V_i(j),0\Big\}\Big\}\;\Big|\;\mathcal{I}_{i1},G\Big]\\
&\geq -\frac{1}{r'}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-r'\max\Big\{\mathbb{E}[e_{i2}\mid \mathcal{I}_{i1}, G+ij, s_{i1}^j]-q-\frac{r}{2}V_i(j),0\Big\}\Big\}\;\Big|\;\mathcal{I}_{i1},G\Big]. \quad (12)
\end{aligned}
$$

Moreover, the inverse-CARA certainty-equivalent operator is increasing in the payoff. Since the signal-contingent optimized payoff under r' is weakly lower than under r,

$$
\begin{aligned}
&-\frac{1}{r'}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-r'\max\Big\{\mathbb{E}[e_{i2}\mid \mathcal{I}_{i1}, G+ij, s_{i1}^j]-q-\frac{r'}{2}V_i(j),0\Big\}\Big\}\;\Big|\;\mathcal{I}_{i1},G\Big]\\
&\leq -\frac{1}{r'}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-r'\max\Big\{\mathbb{E}[e_{i2}\mid \mathcal{I}_{i1}, G+ij, s_{i1}^j]-q-\frac{r}{2}V_i(j),0\Big\}\Big\}\;\Big|\;\mathcal{I}_{i1},G\Big].
\end{aligned}
$$

Combining this inequality with (12) gives $\mathcal{O}_i(ij \mid G; r') \leq \mathcal{O}_i(ij \mid G; r)$. Hence, locally while $M_{i,2}^G = 0$ remains the optimal pre-signal migration choice, $\frac{\partial \mathcal{O}_i(ij|G;r)}{\partial r} \leq 0.$. Given that $\omega_{ij} = O_i(ij \mid G; r)$ in the information-only benchmark, $M_{i,2}^G = 0 \implies \frac{\partial \omega_{ij}}{\partial r} \leq 0.$, The inequality is strict when at least one of the two preceding effects is strict. In particular, it is strict if the signal-contingent optimized payoff is non-degenerate, or if migration occurs after the signal with positive probability and $V_i(j) > 0$.

To build some intuition we can think of this as, when household i would otherwise not migrate, the informational value of the link comes from identifying sufficiently favorable migration opportunities. Greater risk aversion lowers this value through two channels. First, it lowers the certainty-equivalent migration gain associated with every signal realization. Second, even holding those signal-contingent payoffs fixed, greater risk aversion lowers the certainty equivalent of the lottery generated by the possible signal realizations.

At the signal realization at which household i is exactly indifferent between migration and non-migration, the maximum operator has a kink. Since the signal is continuously distributed, this realization occurs with probability zero and does not affect the ex-ante result.

When $M_{i,2}^G = 1$, the same sign result does not follow. In this case, information can be valuable because it allows household i to reverse its baseline migration choice after sufficiently unfavorable signals. Moreover, the baseline migration action itself becomes less valuable as risk aversion increases: $\frac{\partial}{\partial r}\left\{-M_{i,2}^G\left[\mathbb{E}[e_{i2}\mid \mathcal{I}_{i1}, G]-q-\frac{r}{2}V_i\right]\right\} = \frac{1}{2}V_i > 0$. Thus greater risk aversion both lowers the value of signal-contingent migration opportunities and can raise the value of having information that allows household i to avoid the risky baseline migration action. These effects work in opposite directions, so $M_{i,2}^G = 1 \implies \frac{\partial \mathcal{O}_i(ij|G;r)}{\partial r}$ is generally an ambiguous sign.

Combining these results, in the information-only benchmark the value of a link is increasing in the amount of uncertainty that the signal resolves: $\frac{\partial \omega_{ij}}{\partial[V_i - V_i(j)]} > 0$. Moreover, when household i would not migrate without the additional signal, greater absolute risk aversion weakly reduces the value of the link: $M_{i,2}^G = 0$ that leads to $\frac{\partial \omega_{ij}}{\partial r} \leq 0$. When $M_{i,2}^G = 1$, the effect of risk aversion is instead generally ambiguous. This completes the proof of Corollary 3.

A.10 Proof of Proposition 4: Diminishing Informational Returns and Optimal Degree

Gaussian posterior precision is additive across conditionally independent signals. After household i has formed several informative links, posterior uncertainty about z_1 is

$$\left[\frac{1}{V_{i1}} + \sum_{j \in N_{i2} \setminus N_{i1}} \frac{1}{\sigma_{\tilde{e}}^2 + \sigma_\epsilon^2/(1 + d_{j1}^M)}\right]^{-1}.$$

Hence an additional signal of a given precision produces a smaller reduction in posterior variance when household i already possesses more precise information. This follows because, for positive quantities, the difference between the inverse of a given precision and the inverse of that precision plus an additional positive precision becomes smaller as the initial precision increases.

Consider now the welfare value of an additional information link. If household i has already formed $n_i - 1$ new informative links, the value of an additional link ij is

$$\begin{aligned}\mathcal{O}_i(ij \mid G(n_i - 1); r) = & -\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-r \max_{M_{i2} \in \{0,1\}} CE_i\left(M_{i2} \mid \mathcal{I}_{i1}, G(n_i - 1) + ij, s_{i1}^j\right)\Big\} \,\Big|\, \mathcal{I}_{i1}, G(n_i - 1)\Big] \\ & + \frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-rCE_i\left(M_{i,2}^{G(n_i-1)} \mid \mathcal{I}_{i1}, G(n_i - 1) + ij, s_{i1}^j\right)\Big\} \,\Big|\, \mathcal{I}_{i1}, G(n_i - 1)\Big].\end{aligned}$$

If household i already has one additional informative link, the corresponding value of another signal of the same precision is

$$\begin{aligned}\mathcal{O}_i(ij \mid G(n_i); r) = & -\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-r \max_{M_{i2} \in \{0,1\}} CE_i\left(M_{i2} \mid \mathcal{I}_{i1}, G(n_i) + ij, s_{i1}^j\right)\Big\} \,\Big|\, \mathcal{I}_{i1}, G(n_i)\Big] \\ & + \frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\Big[\exp\Big\{-rCE_i\left(M_{i,2}^{G(n_i)} \mid \mathcal{I}_{i1}, G(n_i) + ij, s_{i1}^j\right)\Big\} \,\Big|\, \mathcal{I}_{i1}, G(n_i)\Big].\end{aligned}$$

The difference between these two expressions is the marginal effect of the additional information already contained in $G(n_i)$ relative to $G(n_i - 1)$. Since the additional signal s_{i1}^j is received on top of a more precise information set under $G(n_i)$, it gives a smaller reduction in posterior uncertainty than under $G(n_i - 1)$.

This statistical result alone does not establish $\mathcal{O}_i(ij \mid G(n_i); r) \leq \mathcal{O}_i(ij \mid G(n_i - 1); r)$. The reason is that $\mathcal{O}_i$ measures the value of adapting the subsequent migration decision to the new signal, rather than the reduction in posterior variance itself. Even a signal that reduces uncertainty less may be valuable if the information already available to household i places it close to the migration threshold.

Hence, the proposition imposes the additional condition that the decision value of an additional signal weakly decreases as household i accumulates informative signals. Under this condition, $\mathcal{O}_i(ij \mid G(n_i); r) \leq \mathcal{O}_i(ij \mid G(n_i - 1); r)$. However, for comparable informative links, the marginal informational value of another link weakly decreases as household i acquires more information.

Since an information-only link has gross value $\omega_{ij} = \mathcal{O}_i(ij \mid G; r)$, household i is willing to form an additional informative link only while its marginal informational value weakly exceeds the link cost κ, subject also to the potential partner being willing to form the link.

Accordingly, if the number of new informative links formed by household i, n_i^*, is interior, the last link formed must satisfy $O_i(ij \mid G(n_i - 1); r). \geq \kappa$, whereas the next additional link satisfies $O_i(ij \mid G(n_i); r). < \kappa$.

The first part of the result follows from Gaussian learning: additional signals generate diminishing reductions in posterior variance. The second part additionally requires diminishing decision value, because the welfare value $\mathcal{O}_i(ij \mid G; r)$ depends on whether the additional information changes the subsequent migration decision.

A.11 Proof of Proposition 5: Link Formation with Information and Income Sharing

We now return to the full model. Conditional on the period-2 migration choices of households other than i, household i's period-2 consumption under the network being evaluated is

$$c_{i2} = \frac{y_{i2} + e_{i2} M_{i2}}{d_{i2} + 1} + \sum_{k \in N_{i2}} \frac{y_{k2} + e_{k2} M_{k2}}{d_{k2} + 1} - q M_{i2}. \tag{13}$$

If j is the new peer, the sum contains $\frac{y_{j2} + e_{j2} M_{j2}}{d_{j2} + 1}$. Thus, if $M_{j2} = 1$, household i receives a share $e_{j2}/(d_{j2} + 1)$ of household j's urban income.

Now to compute the conditional mean we use $\mathbb{E}[y_{k2}] = \bar{y}$ and $\mathbb{E}[e_{k2} \mid \mathcal{I}_{i2}] = \rho \widehat{\mu}_{z,i,2}$ for migration income with the common component, the conditional mean is (and it is later use

in the next result):

$$\mathbb{E}[c_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^{j}, \{M_{k2}\}_k] = \frac{\bar{y}}{d_{i2}+1} + \sum_{k \in N_{i2}} \frac{\bar{y}}{d_{k2}+1} + \rho \widehat{\mu}_{z,i,2} \left(\frac{M_{i2}}{d_{i2}+1} + \sum_{k \in N_{i2}} \frac{M_{k2}}{d_{k2}+1} \right) - qM_{i2}. \tag{14}$$

The term $\rho \widehat{\mu}_{z,i,2} \sum_{k \in N_{i2}} \frac{M_{k2}}{d_{k2}+1}$ is the expected migration income that household i receives from migrating peers. It is present even when $M_{i2} = 0$.

For the conditional variance and covariance terms, notice that the rural-income shocks are independent across households. Their contribution to the variance is

$$\frac{\sigma^2}{(d_{i2}+1)^2} + \sigma^2 \sum_{k \in N_{i2}} \frac{1}{(d_{k2}+1)^2}.$$

Urban income has a common component and an idiosyncratic component. Conditional on the signal, the uncertain common period-2 component has variance $\rho^2 \widehat{\sigma}_{z,i,2}^2 + \sigma_\lambda^2$. Its coefficient in household i's consumption is $\frac{M_{i2}}{d_{i2}+1} + \sum_{k \in N_{i2}} \frac{M_{k2}}{d_{k2}+1}$. Therefore its variance contribution is

$$\left(\rho^2 \widehat{\sigma}_{z,i,2}^2 + \sigma_\lambda^2\right) \left(\frac{M_{i2}}{d_{i2}+1} + \sum_{k \in N_{i2}} \frac{M_{k2}}{d_{k2}+1} \right)^2.$$

This square already contains every covariance generated by the common urban-income component. For example, the covariance between household i's own migration income and the migration income received from a migrating peer k appears as the cross term

$$2\left(\rho^2 \widehat{\sigma}_{z,i,2}^2 + \sigma_\lambda^2\right) \frac{M_{i2}}{d_{i2}+1} \frac{M_{k2}}{d_{k2}+1}.$$

The idiosyncratic urban-income shocks are independent across households, so they do not contain cross terms $\sigma_{\tilde{e}}^2 \left(\frac{M_{i2}}{(d_{i2}+1)^2} + \sum_{k \in N_{i2}} \frac{M_{k2}}{(d_{k2}+1)^2} \right)$. Combining the components gives the conditional variable used in next results:

$$\begin{aligned} \mathrm{Var}(c_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^{j}, \{M_{k2}\}_k) &= \frac{\sigma^2}{(d_{i2}+1)^2} + \sigma^2 \sum_{k \in N_{i2}} \frac{1}{(d_{k2}+1)^2} \\ &+ \left(\rho^2 \widehat{\sigma}_{z,i,2}^2 + \sigma_\lambda^2\right) \left(\frac{M_{i2}}{d_{i2}+1} + \sum_{k \in N_{i2}} \frac{M_{k2}}{d_{k2}+1} \right)^2 + \sigma_{\tilde{e}}^2 \left(\frac{M_{i2}}{(d_{i2}+1)^2} + \sum_{k \in N_{i2}} \frac{M_{k2}}{(d_{k2}+1)^2} \right). \end{aligned}$$

Conditional gain from i household migration is calculating by subtracting the certainty equivalent under $M_{i2} = 0$ from the certainty equivalent under $M_{i2} = 1$.

From (14), the rural-income terms cancel. The expected migration income received from peers also cancels because $\rho\widehat{\mu}_{z,i,2}\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1}$ appears under both $M_{i2}=0$ and $M_{i2}=1$. The mean difference is therefore $\frac{\rho\widehat{\mu}_{z,i,2}}{d_{i2}+1}-q$. For the common-risk term, use

$$\left(\frac{1}{d_{i2}+1}+\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1}\right)^2-\left(\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1}\right)^2=\frac{2}{d_{i2}+1}\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1}+\frac{1}{(d_{i2}+1)^2}.$$

The first term on the right-hand side is the additional covariance exposure between household i's own urban income and the urban income it already receives from migrating peers. The second term is the variance of household i's own exposure to the common migration shock.

For the idiosyncratic urban-income component, switching from $M_{i2}=0$ to $M_{i2}=1$ adds only $\frac{\sigma_{\tilde{e}}^2}{(d_{i2}+1)^2}$. Hence the conditional certainty-equivalent gain from household i's own migration is

$$\begin{aligned}&CE_i\big(M_{i2}=1\mid\mathcal{I}_{i1},G+ij,s_{i1}^j,\{M_{k2}\}_{k\neq i}\big)-CE_i\big(M_{i2}=0\mid\mathcal{I}_{i1},G+ij,s_{i1}^j,\{M_{k2}\}_{k\neq i}\big)\\&=\frac{\rho\widehat{\mu}_{z,i,2}}{d_{i2}+1}-q-\frac{r}{2}\Big[\left(\rho^2\widehat{\sigma}_{z,i,2}^2+\sigma_\lambda^2\right)\left(\frac{2}{d_{i2}+1}\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1}+\frac{1}{(d_{i2}+1)^2}\right)+\frac{\sigma_{\tilde{e}}^2}{(d_{i2}+1)^2}\Big].\end{aligned}\tag{15}$$

This expression determines whether household i exercises its own period-2 migration option after observing the signal. Peers' expected migration income cancels from the mean comparison, but peers' migration still affects the relative attractiveness of household i's own migration through the common-risk covariance term.

A.12 Full-Model Option Value with Peer Migration Income Included

The option value refers to the option to adjust household i's own migration decision M_{i2}. The payoffs associated with $M_{i2}=0$ and $M_{i2}=1$, however, contain the rural and urban income received from peers. Therefore peer migration income must be included inside the action-specific certainty equivalents that enter $\mathcal{O}_i$; it is not an additional term added outside the option value. The comparison is conditional on the continuation migration choices of the other households, which and they are fixed when household i chooses its own migration decision. Substituting the conditional certainty equivalents derived above into the ex-ante

CARA expression, the informational option value can be written as:

$$\begin{aligned}\mathcal{O}_i(ij \mid G; r) = & -\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r \max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j)\right\} \mid \mathcal{I}_{i1}, G\right] \\ & + \frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-rCE_i(M_{i,2}^G \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j)\right\} \mid \mathcal{I}_{i1}, G\right].\end{aligned} \tag{16}$$

The first term evaluates the optimal migration decision after observing the additional signal, and the second term evaluates the baseline migration decision under the same post-link network, so you use expressions 14 and A.11 and substitute for M^G for the second term. The expected migration income from peers enters both terms through $\rho\widehat{\mu}_{z,i,2}\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1}$, as included in the conditional certainty equivalents. If j is the new peer and $M_{j2}=1$, this includes the expected urban-income share $\frac{\rho\widehat{\mu}_{z,i,2}}{d_{j2}+1}$. The conditional certainty equivalents also account for the covariance exposure to migrating peers through the squared common-risk term.

Equation (16) has exactly the same economic definition as the information-only option value: it is the value of allowing i to choose M_{i2} after observing the new signal. The closed form is different because, with income sharing, even the payoff from $M_{i2}=0$ can depend on the signal through urban income received from migrating peers. Hence the simple information-only truncated-normal expression cannot generally be reused for the full model.

At the link-formation stage, the other households' period-2 migration choices have not yet been realized. If those choices depend on subsequent signals, the ex-ante value of the link integrates (16) over the migration profiles induced by equilibrium continuation strategies. Because migration choices enter quadratically in the common-risk term, a future M_{k2} cannot in general be replaced by its unconditional probability inside the variance expression.

A.13 Full-Model Link-Value Decomposition and Proposition 5

The net value to household i of adding link ij is $\Delta CE_i(ij \mid G) = CE_i(G+ij) - CE_i(G) - \kappa$.

We distinguish the financial effect of the link from the value of optimizing again, hold household i's migration policy fixed at the policy that would be followed under G, denoted by $M_{i,2}^G$. For a fixed policy, pure information has no ex-ante value by the law of iterated expectations. The difference between $G+ij$ and G then comes from the change in the consumption lottery generated by the new sharing relationship.

Under CARA preferences, the exact ex-ante certainty equivalent of any consumption lottery is $CE_i = -\frac{1}{r}\log\mathbb{E}\left[\exp(-rc_{i2})\right]$. Therefore, holding the baseline migration policy fixed, the financial effect of the new relationship is $CE_i(G+ij; M_{i,2}^G) - CE_i(G; M_{i,2}^G)$.

Allowing household i to reconsider its migration decision after observing the new signal

adds the option value $\mathcal{O}_i(ij \mid G; r)$. Hence,

$$\Delta CE_i(ij \mid G) = CE_i(G + ij; M_{i,2}^G) - CE_i(G; M_{i,2}^G) + \mathcal{O}_i(ij \mid G; r) - \kappa.$$

This decomposition is exact when the other households' continuation behavior is held fixed across the comparison. The financial component is evaluated using the ex-ante certainty equivalent of the actual consumption lottery under the baseline migration policy.

When consumption is normally distributed under a fixed migration action and migration profile, the financial component can be written as $\Delta E[c_{i2}] - \frac{r}{2}\Delta \operatorname{Var}(c_{i2})$. More generally, when migration decisions depend on realized signals, the ex-ante consumption distribution might be not normal, and the exact CARA certainty-equivalent expression must be used.

The posterior-variance reduction generated by the signal should not be added separately, and its contribution is already in the option value through the ability to change migration after observing the signal. The three economic channels are:

1. Changes in expected consumption: redistribution generated by the new relationship;
2. Changes in consumption risk: insurance gains or additional financial risk generated by sharing;
3. $\mathcal{O}_i(ij \mid G; r)$: the value of adapting household i's own migration decision to the additional information.

Applying the same comparison to household j and imposing bilateral consent gives Proposition 5. Under the tie-breaking rule, link ij forms if and only if

$$\Delta CE_i(ij \mid G) \geq 0, \qquad \Delta CE_j(ij \mid G) \geq 0, \qquad \max\{\Delta CE_i(ij \mid G), \Delta CE_j(ij \mid G)\} > 0.$$

A.14 Comparative Statics in the Full Model

A.14.1 Signal precision

Hold the prior distribution, network, sharing rule, income distributions, and the other households' migration choices fixed. A more precise signal is a improvement of the information available to household i. For any fixed M_{i2}, the household can ignore the additional precision and reproduce the same action and expected utility as before. Allowing the action to respond optimally to a more informative signal therefore cannot lower ex-ante expected utility. Hence, greater signal precision leads to $O_i(ij \mid G; r)$ weakly increases. The inequality is strict when the extra precision changes household i's optimal migration action for a set of signal realizations with positive probability.

Conditional on j having migrated in period 1, the model's signal technology implies that an increase in d_{j1}^M decreases $\sigma_{\tilde{e}}^2 + \frac{\sigma_\epsilon^2}{1+d_{j1}^M}$ and also increases $\mathcal{O}_i(ij \mid G; r) \uparrow$. If the financial side (resources) of the link are held fixed, the total link value therefore weakly rises with signal precision.

A.14.2 Prior uncertainty

Let $V_{i1} = \mathrm{Var}(z_1 \mid \mathcal{I}_{i1}, G)$ measure prior uncertainty about the learnable common component. Hold the conditional mean, the signal technology, and the migration profile fixed. Expected consumption does not depend directly on V_{i1}, while the common-risk component of ex-ante consumption variance is proportional to $(\rho^2 V_{i1} + \sigma_\lambda^2)\left(\frac{M_{i,2}^G}{d_{i2}+1} + \sum_{k \in N_{i2}} \frac{M_{k2}}{d_{k2}+1}\right)^2$. Therefore, under fixed migration actions and a fixed continuation migration profile, the financial component can be evaluated using the Gaussian certainty equivalent. In this conditional comparison, we obtain

$$\frac{\partial \Delta CE_i(ij \mid G)}{\partial V_{i1}} = -\frac{r\rho^2}{2}\Delta\left[\left(\frac{M_{i,2}^G}{d_{i2}+1} + \sum_{k \in N_{i2}} \frac{M_{k2}}{d_{k2}+1}\right)^2\right] + \frac{\partial \mathcal{O}_i(ij \mid G; r)}{\partial V_{i1}}.$$

Although the marginal link is ij, the common-risk component depends on all migrating peers because uncertainty about z_1 affects their urban incomes jointly; taking the difference between $G + ij$ and G cancels the unchanged components and retains only the additional variance and covariance exposure generated by the new link.

The first term is positive when the new sharing relationship lowers household i's exposure to the common migration shock and negative when the link raises that exposure. The second term captures how additional prior uncertainty changes the value of being able to revise M_{i2} after observing the signal. Unlike in the information-only special case with baseline migration, this option-value effect is not generally enough to sign the total derivative. Hence the effect of greater prior uncertainty on the value of an information-and-financial link is generally ambiguous.

Then, greater prior uncertainty increases the amount that can potentially be learned, but it also makes common urban-income exposure more risky. A financial link can either diversify that exposure or add to it, depending on the sharing weights and the migration choices of the linked households.

A.14.3 Absolute risk aversion

Within a region in which $M_{i,2}^G$ does not change and holding the income distributions fixed, differentiate the full link-value decomposition:

$$\frac{\partial \Delta CE_i(ij \mid G)}{\partial r} = \frac{\partial}{\partial r}\left[CE_i(G+ij; M_{i,2}^G) - CE_i(G; M_{i,2}^G)\right] + \frac{\partial \mathcal{O}_i(ij \mid G; r)}{\partial r}.$$

The first term is positive if the new financial relationship reduces actual consumption variance and negative if it increases actual consumption variance.

For any fixed migration action, CARA-normality gives

$$\frac{\partial}{\partial r} CE_i\big(M_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j, \{M_{k2}\}_{k\neq i}\big) = -\frac{1}{2}\operatorname{Var}\big(c_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j, M_{i2}, \{M_{k2}\}_{k\neq i}\big).$$

Applying this result to the migration-gain expression (15) gives

$$\begin{aligned}
&\frac{\partial}{\partial r}\Big[CE_i(M_{i2}=1 \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j, \{M_{k2}\}_{k\neq i}) - CE_i(M_{i2}=0 \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j, \{M_{k2}\}_{k\neq i})\Big] \\
&= -\frac{1}{2}\Big[\left(\rho^2 \widehat{\sigma}_{z,i,2}^2 + \sigma_\lambda^2\right)\left(\frac{2}{d_{i2}+1}\sum_{k\in N_{i2}}\frac{M_{k2}}{d_{k2}+1} + \frac{1}{(d_{i2}+1)^2}\right) + \frac{\sigma_e^2}{(d_{i2}+1)^2}\Big] < 0.
\end{aligned}$$

Thus greater risk aversion lowers the conditional certainty-equivalent gain from household i's own migration for every signal realization. It shifts the migration cutoff against migration.

This threshold effect does not sign $\partial \mathcal{O}_i/\partial r$. The option value is an ex-ante inverse-CARA certainty equivalent, so changing r changes both the set of signal realizations under which migration is optimal and the way the resulting signal-contingent utilities are aggregated. The case with information-only above demonstrates this ambiguity. Hence $\frac{\partial \mathcal{O}_i(ij|G;r)}{\partial r}$ is generally ambiguous, and therefore $\frac{\partial \Delta CE_i(ij|G)}{\partial r}$ is also generally ambiguous. An special case is one in which the signal never changes household i's migration action. Then $\mathcal{O}_i = 0$ and $\partial \mathcal{O}_i/\partial r = 0$, so $\frac{\partial \Delta CE_i(ij|G)}{\partial r} = -\frac{1}{2}\Delta \operatorname{Var}(c_{i2})$. The derivative is positive for a link that reduces consumption variance and negative for a link that increases consumption variance. At values of r where $M_{i,2}^G$ changes, the link-value function might have a kink.

A.15 Proof of Proposition 6: Strategic Substitutability in Link Formation

Consider households i and k, each of which may form a new link with the same potential partner j. To establish strategic substitutability, we compare the marginal value to household i of forming ij when kj is absent with its marginal value when kj is already present. Strategic

substitutability requires $\Delta CE_i(ij \mid G + kj) < \Delta CE_i(ij \mid G)$.

Take d_{i2} and d_{j2} to be the period-2 degrees in G before either candidate link is added. If ij forms while kj is absent, household i's sharing denominator becomes $d_{i2}+2$ and household j's sharing denominator becomes $d_{j2}+2$. If kj has already formed when ij is added, household i still has denominator $d_{i2}+2$, whereas household j now has denominator $d_{j2}+3$. Hence

$$\frac{1}{d_{j2}+3} < \frac{1}{d_{j2}+2}.$$

Formation of kj therefore reduces the share of j's income received by i through a subsequent link ij. At the same time, it reduces household i's exposure to the risk associated with j's income.

Using the link-value decomposition established above,

$$\begin{aligned}\Delta CE_i(ij \mid G + kj) - \Delta CE_i(ij \mid G) = [CE_i(G + kj + ij; M_{i,2}^{G+kj}) - CE_i(G + kj; M_{i,2}^{G+kj})] \\ - [CE_i(G + ij; M_{i,2}^{G}) - CE_i(G; M_{i,2}^{G})] + \mathcal{O}_i(ij \mid G + kj; r) - \mathcal{O}_i(ij \mid G; r).\end{aligned} \tag{17}$$

The link-formation cost cancels because it is the same in the two comparisons. The first two expressions capture the change in the financial value of forming link ij, evaluated using exact CARA certainty equivalents. The remaining terms capture the change in the informational option value.

The comparison can be disentangled into three channels. First, the higher degree of j changes the certainty-equivalent contribution of j's own rural and, when $M_{j2} = 1$, urban income. Second, when j migrates, the lower sharing weight reduces the common-shock covariance between j's urban income and the other migration income entering household i's consumption. Third, kj does not change the information transmitted by j, but it can change the option value by changing the payoff comparison between household i's subsequent migration choices.

The option value when kj is absent is

$$\begin{aligned}\mathcal{O}_i(ij \mid G; r) = & -\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r \max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j)\right\} \mid \mathcal{I}_{i1}, G\right] \\ & + \frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-rCE_i(M_{i,2}^{G} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j)\right\} \mid \mathcal{I}_{i1}, G\right].\end{aligned} \tag{18}$$

When kj is already present,

$$\mathcal{O}_i(ij \mid G+kj; r) = -\frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-r \max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid \mathcal{I}_{i1}, G+kj+ij, s_{i1}^j)\right\} \mid \mathcal{I}_{i1}, G+kj\right] + \frac{1}{r}\log \mathbb{E}_{s_{i1}^j}\left[\exp\left\{-rCE_i(M_{i,2}^{G+kj} \mid \mathcal{I}_{i1}, G+kj+ij, s_{i1}^j)\right\} \mid \mathcal{I}_{i1}, G+kj\right]. \tag{19}$$

A link kj does not change the realization or precision of s_{i1}^j. Its precision is determined by household j's period-1 migration outcome and d_{j1}^M, both fixed before period-2 links are formed. So, the information about urban returns provided by j is the same in (18) and (19). Any difference between the two option values will arise from a change in the payoff from acting on the signal rather than from a change in the information itself.

We now consider separately the cases $M_{j2}=0$ and $M_{j2}=1$.

Case 1: $M_{j2}=0$. When $M_{j2}=0$, household j contributes no period-2 urban income to household i. Hence all terms involving j's urban income disappear. The only direct effect of the prior formation of kj on the value of a subsequent link ij is therefore that household j's rural income is divided among one additional neighbor.

If ij forms while kj is absent, the certainty-equivalent contribution of j's rural income to household i is $\frac{\bar{y}}{d_{j2}+2} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+2)^2}$. If kj is already present when ij forms, the corresponding contribution is $\frac{\bar{y}}{d_{j2}+3} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+3)^2}$. Therefore, for either $M_{i2}\in\{0,1\}$ and every realization of s_{i1}^j,

$$\begin{aligned} &CE_i(M_{i2} \mid \mathcal{I}_{i1}, G+kj+ij, s_{i1}^j) - CE_i(M_{i2} \mid \mathcal{I}_{i1}, G+ij, s_{i1}^j) \\ &= \left[\frac{\bar{y}}{d_{j2}+3} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+3)^2}\right] - \left[\frac{\bar{y}}{d_{j2}+2} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+2)^2}\right]. \end{aligned} \tag{20}$$

Importantly, the right-hand side of (20) does not depend on M_{i2} or on s_{i1}^j. Formation of kj therefore shifts the certainty equivalent associated with $M_{i2}=0$ and $M_{i2}=1$ by exactly the same amount for every signal realization. Hence it does not change household i's signal-contingent migration comparison.

Moreover, before ij is formed, link kj does not create a direct sharing relationship involving household i. Under the maintained comparison, holding the continuation choices of the other households fixed, household i therefore faces the same pre-ij migration problem and $M_{i,2}^{G+kj} = M_{i,2}^G$.

To see directly why the option value is unchanged, the common shift in (20) enters both

action-specific certainty equivalents. Thus,

$$\max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid \mathcal{I}_{i1}, G + kj + ij, s_{i1}^j) = \max_{M_{i2}\in\{0,1\}} CE_i(M_{i2} \mid \mathcal{I}_{i1}, G + ij, s_{i1}^j)$$
$$+ \left[\frac{\bar{y}}{d_{j2}+3} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+3)^2}\right] - \left[\frac{\bar{y}}{d_{j2}+2} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+2)^2}\right].$$

The same shift appears in the certainty equivalent associated with the baseline action. Under CARA utility, it therefore factors out of both expectations defining the option value as the same multiplicative exponential term and cancels after taking logarithms. Consequently,

$$M_{j2} = 0 \quad \Longrightarrow \quad \mathcal{O}_i(ij \mid G + kj; r) = \mathcal{O}_i(ij \mid G; r). \tag{21}$$

Hence, when $M_{j2} = 0$, the option-value channel is exactly zero and the total effect of kj on the marginal value of ij is the change in the certainty equivalent of j's rural-income contribution:

$$\begin{aligned} &\Delta CE_i(ij \mid G + kj) - \Delta CE_i(ij \mid G) \\ &= \left[\frac{\bar{y}}{d_{j2}+3} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+3)^2}\right] - \left[\frac{\bar{y}}{d_{j2}+2} - \frac{r}{2}\frac{\sigma^2}{(d_{j2}+2)^2}\right] \\ &= -\frac{\bar{y}}{(d_{j2}+2)(d_{j2}+3)} + \frac{r\sigma^2}{2}\frac{2d_{j2}+5}{(d_{j2}+2)^2(d_{j2}+3)^2}. \end{aligned} \tag{22}$$

The first term in (22) is negative because the higher degree of j reduces the expected rural income received by i through ij. The second term is positive because the same reduction in the sharing weight lowers household i's exposure to j's rural-income risk.

Assumption 2 guarantees that the expected-income loss dominates the reduction in risk. In particular,

$$CE\left(\frac{y_{it}}{d_{it}+1}\right) = \frac{\bar{y}}{d_{it}+1} - \frac{r}{2}\frac{\sigma^2}{(d_{it}+1)^2} > 0 \qquad \forall d_{it} \geq 0.$$

Taking $d_{it} = 0$ implies $2\bar{y} > r\sigma^2$. Furthermore,

$$(d_{j2}+2)(d_{j2}+3) - (2d_{j2}+5) = d_{j2}^2 + 3d_{j2} + 1 > 0.$$

Therefore,

$$2\bar{y}(d_{j2}+2)(d_{j2}+3) > r\sigma^2(d_{j2}+2)(d_{j2}+3) > r\sigma^2(2d_{j2}+5).$$

This is precisely the condition required for the right-hand side of (22) to be strictly negative. Hence

$$M_{j2} = 0 \quad \Longrightarrow \quad \Delta CE_i(ij \mid G + kj) < \Delta CE_i(ij \mid G). \tag{23}$$

Thus, when household j does not migrate in period 2, links ij and kj are strategic substitutes. Formation of kj does not change the informational option value of ij, while the positive rural-income certainty-equivalent assumption guarantees that the reduction in the expected rural-income share received from j dominates the benefit from the corresponding reduction in rural-income risk.

Case 2: $M_{j2} = 1$.

When $M_{j2} = 1$, the negative rural-income effect established in Case 1 remains present. In addition, household i receives a share of household j's urban income through link ij. Formation of kj reduces this share because j's sharing denominator increases from $d_{j2} + 2$ to $d_{j2} + 3$.

Conditional on the information available after the signal is observed, the mean and variance of household j's urban income from the perspective of household i are

$$\begin{aligned} \mathbb{E}[e_{j2} \mid \mathcal{I}_{i1}, s^j_{i1}] &= \rho \widehat{\mu}_{z,i,2}, \\ \mathrm{Var}(e_{j2} \mid \mathcal{I}_{i1}, s^j_{i1}) &= \rho^2 \widehat{\sigma}^2_{z,i,2} + \sigma^2_\lambda + \sigma^2_{\widetilde{e}}. \end{aligned}$$

Hence the certainty-equivalent contribution of household j's own urban income through ij is $\frac{\rho\widehat{\mu}_{z,i,2}}{d_{j2}+2} - \frac{r}{2}\frac{\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}}{(d_{j2}+2)^2}$, when kj is absent, and $\frac{\rho\widehat{\mu}_{z,i,2}}{d_{j2}+3} - \frac{r}{2}\frac{\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}}{(d_{j2}+3)^2}$ when kj is present.

The difference is therefore

$$\begin{aligned} &\left[\frac{\rho\widehat{\mu}_{z,i,2}}{d_{j2}+3} - \frac{r}{2}\frac{\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}}{(d_{j2}+3)^2}\right] - \left[\frac{\rho\widehat{\mu}_{z,i,2}}{d_{j2}+2} - \frac{r}{2}\frac{\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}}{(d_{j2}+2)^2}\right] \\ &= -\frac{\rho\widehat{\mu}_{z,i,2}}{(d_{j2}+2)(d_{j2}+3)} + \frac{r}{2}\left(\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}\right)\frac{2d_{j2}+5}{(d_{j2}+2)^2(d_{j2}+3)^2}. \end{aligned} \tag{24}$$

The first term in (24) is the loss in expected urban income received from j, whereas the second is the benefit from the reduction in the variance of j's own urban-income contribution. Therefore, the own urban-income effect is negative if and only if

$$2\rho\widehat{\mu}_{z,i,2}(d_{j2}+2)(d_{j2}+3) > r\left(\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda+\sigma^2_{\widetilde{e}}\right)(2d_{j2}+5). \tag{25}$$

The fact that $M_{j2} = 1$ means that migration is optimal for household j under household j's own information. In particular, j would not migrate if the certainty-equivalent gain

from its own migration were negative. This restriction implies that the urban opportunity is sufficiently attractive from household j's perspective. However, it does not imply (25), because the urban-income contribution in (24) is evaluated using household i's information about j's income. Moreover, positivity of a certainty equivalent at a given sharing weight does not alone imply that reducing that weight necessarily lowers the certainty equivalent: the accompanying reduction in risk may be valuable. Condition (25) gives the exact requirement for the expected-income sharing (dilution) to dominate this own-risk reduction.

A useful special case arises when neither household i or any of its period-1 neighbors migrated. In that case, household i has not observed any migration-income realization with which to update its beliefs about urban returns, so the mean and variance entering (25) are the corresponding prior moments. Under the maintained assumption that the prior certainty equivalent of urban income is positive, the resulting expected-income net of own risk is positive, which simplifies condition (25).

There is also a separate effect through the common component of urban income. Household j's urban income is correlated with the other migration income entering household i's consumption. When kj forms, the sharing weight on j falls from $1/(d_{j2}+2)$ to $1/(d_{j2}+3)$. Holding household i's migration action and the migration choices of the other households fixed, the change in the covariance component of household i's certainty equivalent is

$$-\,r\left(\rho^2\widehat{\sigma}^2_{z,i,2}+\sigma^2_\lambda\right)\left[\frac{M_{i2}}{d_{i2}+2}+\sum_{\ell\in N_{i2}}\frac{M_{\ell 2}}{d_{\ell 2}+1}\right]\left(\frac{1}{d_{j2}+3}-\frac{1}{d_{j2}+2}\right)>0, \tag{26}$$

whenever household i is already exposed to migration income through itself or another migrating neighbor. The inequality follows because $\frac{1}{d_{j2}+3}-\frac{1}{d_{j2}+2}<0$. Thus formation of kj reduces the covariance risk transmitted through j, which raises household i's certainty equivalent. This effect works against strategic substitutability.

Formation of kj can also change the option value of ij. To see why, compare household i's signal-contingent gain from its own migration after ij has formed. The expected urban-income contribution received from j appears under both $M_{i2}=0$ and $M_{i2}=1$ and therefore cancels from the conditional mean migration comparison. The difference between the two

networks instead arises from the common-risk covariance term:

$$\begin{aligned}
&[CE_i(M_{i2}=1 \mid G+kj+ij, s_{i1}^j) - CE_i(M_{i2}=0 \mid G+kj+ij, s_{i1}^j)] \\
&- [CE_i(M_{i2}=1 \mid G+ij, s_{i1}^j) - CE_i(M_{i2}=0 \mid G+ij, s_{i1}^j)] \\
=& -\frac{r}{2}\left(\rho^2\widehat{\sigma}_{z,i,2}^2+\sigma_\lambda^2\right)\frac{2}{d_{i2}+2}\left(\frac{1}{d_{j2}+3}-\frac{1}{d_{j2}+2}\right) \\
=& \frac{r\left(\rho^2\widehat{\sigma}_{z,i,2}^2+\sigma_\lambda^2\right)}{(d_{i2}+2)(d_{j2}+2)(d_{j2}+3)} > 0. \qquad (27)
\end{aligned}$$

Hence, conditional on ij being present, formation of kj makes household i's own migration relatively more attractive. The reason is that the higher degree of j reduces the common-risk covariance borne by household i when it also migrates.

Equation (27) does not imply that link ij becomes more valuable. It only shows that kj changes household i's signal-contingent migration comparison. Since the option value is generated by household i's ability to adapt its migration decision after observing s_{i1}^j, this change can alter the migration threshold and therefore $\mathcal{O}_i(ij \mid G+kj; r) - \mathcal{O}_i(ij \mid G; r)$. The sign of this option-value difference is not determined in general.

Thus, conditional on $M_{j2}=1$, the effect of kj on the marginal value of ij can be disentangled into four components:

$$\begin{aligned}
\Delta CE_i(ij \mid G+kj) - \Delta CE_i(ij \mid G) = &\underbrace{\text{rural-income effect}}_{<0} \\
&+ \underbrace{\text{own urban-income effect}}_{\text{negative under (25)}} \\
&+ \underbrace{\text{reduction in common-risk covariance}}_{\geq 0} \\
&+ \underbrace{\mathcal{O}_i(ij \mid G+kj; r) - \mathcal{O}_i(ij \mid G; r)}_{\text{not signed}}.
\end{aligned}$$

The rural-income effect is strictly negative by Assumption 2. The own urban-income effect is negative whenever the expected-income dilution condition (25) holds. The reduction in common-risk covariance raises the value of the link, while the change in the option value is not signed in general.

Therefore, when $M_{j2}=1$, potential links ij and kj are strategic substitutes whenever the negative rural-income effect, together with any negative urban-income dilution effect, dominates the benefit from lower common-risk covariance and any offsetting increase in the

option value. Equivalently,

$$\begin{aligned} 0 > [CE_i(G + kj + ij; M_{i,2}^{G+kj}) - CE_i(G + kj; M_{i,2}^{G+kj})] \\ -[CE_i(G + ij; M_{i,2}^{G}) - CE_i(G; M_{i,2}^{G})] + \mathcal{O}_i(ij \mid G + kj; r) - \mathcal{O}_i(ij \mid G; r). \end{aligned} \tag{28}$$

Under this condition, $\Delta CE_i(ij \mid G + kj) < \Delta CE_i(ij \mid G)$.

The first two expressions compare the exact financial certainty-equivalent effects of forming link ij with and without the prior formation of kj. Under fixed migration actions and a fixed continuation migration profile, these financial effects can be decomposed into expected-income dilution and changes in consumption risk, as derived above. The last two terms capture the difference in the informational option value. Since the higher degree of household j changes the risk associated with household i's own migration, the option-value difference is not generally signed. Then, strategic substitutability holds when the negative financial effects dominate any outweighing increase in the informational option value.

The two cases therefore deliver distinct conclusions. When $M_{j2} = 0$, the option-value difference is zero and the positive-rural-certainty-equivalent assumption is sufficient to imply strategic substitutability. When $M_{j2} = 1$, both the rural and the own urban certainty-equivalent contributions of j fall with j's degree under the maintained positive-certainty-equivalent conditions. However, the higher degree of j also reduces common-shock covariance risk and can change the option value. Strategic substitutability therefore follows when the two income-dilution effects dominate these offsetting channels.

The calculations above condition on a given continuation migration profile of the other households. At the link-formation stage, if period-2 migration choices are themselves determined by subsequent signals, the ex-ante link value integrates over the continuation profiles induced by equilibrium strategies. The same strategic-substitutability conclusion follows if the stated inequality holds for the relevant continuation profiles, or equivalently after taking the corresponding ex-ante expectation.

Finally, in the information-only benchmark the financial-sharing terms are absent. Formation of kj does not change the realization or precision of the period-1 signal transmitted by j, and therefore does not change the informational value of ij. The strategic-substitution mechanism characterized above is consequently generated by the income-sharing relationship.

A.16 Proof of Corollary 4: Characteristics of Attractive Partners

The two dimensions of household j's network have distinct effects.

First, conditional on j having migrated in period 1, greater migration exposure among

j's neighbors increases the precision of the signal transmitted by j because

$$\frac{\partial}{\partial d_{j1}^M}\left(\sigma_{\tilde{e}}^2+\frac{\sigma_\epsilon^2}{1+d_{j1}^M}\right)<0.$$

As proved above, greater signal precision raises the option value. Hence a household with greater migration exposure is a more attractive information provider, holding the financial consequences of the relationship fixed.

Second, a larger total degree of j reduces the income share received by a new partner. Holding migration exposure fixed, a larger d_{j2} lowers $\frac{\mathbb{E}[y_{j2}+e_{j2}M_{j2}]}{d_{j2}+1}$. It also lowers the risk transmitted through j because the variance contribution is scaled by $1/(d_{j2}+1)^2$. Under the dominance condition in Proposition 6, the expected-income sharing dominates the reduction in risk exposure and any offsetting option-value effect, so the value of linking to j decreases with j's total degree.

Finally, holding information and risk effects fixed, a larger expected net transfer from household i to household j lowers $\Delta\mathbb{E}[c_{i2}]$ and therefore lowers the value of the link. These observations establish Corollary 4.

A.17 Proof of Corollary 5: Marginal Value of Additional Links

The informational part inherits the qualification from Proposition 4. Gaussian updating implies that each additional signal produces a smaller reduction in posterior variance as household i accumulates information. Under the additional condition that the decision value of successive signals weakly declines as information becomes more precise, the marginal informational contribution of another link also declines.

The financial component of another link, $\Delta\mathbb{E}[c_{i2}]-\frac{r}{2}\Delta\operatorname{Var}(c_{i2})$, has no fixed sign. An additional relationship may increase expected consumption and improve diversification, or it may lower expected consumption through redistribution and expose the household to additional income risk. Consequently, introducing transfers can either increase or decrease the optimal number of links relative to the information-only benchmark. Therefore, the full-model network may be either denser or sparser than the information-only network.

A.18 Proof of Corollary 6: Risk Aversion and Link Formation

The full-model derivative is

$$\frac{\partial\Delta CE_i(ij\mid G)}{\partial r}=\frac{\partial}{\partial r}\left[CE_i(G+ij;M_{i,2}^G)-CE_i(G;M_{i,2}^G)\right]+\frac{\partial\mathcal{O}_i(ij\mid G;r)}{\partial r}.$$

The first term captures how risk aversion changes the financial value of the new relationship, while the second captures its effect on the informational option value. When migration actions and the continuation migration profile are fixed, consumption is normally distributed and the derivative of the financial component reduces to $-\frac{1}{2}\Delta \operatorname{Var}(c_{i2})$. In this conditional case, the financial effect is positive when the link reduces consumption variance and negative when it increases consumption variance.

For the information-only benchmark, the option-value comparative static was derived above. Since $\omega_{ij} = \mathcal{O}_i(ij \mid G; r)$, $M^G_{i,2} = 0 \implies \frac{\partial \omega_{ij}}{\partial r} \leq 0$, whereas $M^G_{i,2} = 1 \implies \frac{\partial \omega_{ij}}{\partial r}$ is generally ambiguous. These signs do not directly extend to the option-value component of the full model. Once the new link also creates an income-sharing relationship, the signal can affect the value of consumption received through that relationship as well as household i's own migration decision. Consequently, $\partial \mathcal{O}_i(ij \mid G; r)/\partial r$ is not generally signed in the full model.

Combining this with the potentially positive or negative financial risk-sharing term implies that the effect of greater absolute risk aversion on full-model link value is generally ambiguous, including conditional on $M^G_{i,2}$.

A.19 Equilibrium Interpretation

The theoretical analysis is conditional on a period-2 network satisfying the bilateral-consent conditions stated in Section 2.2. For every candidate link ij, link values are evaluated taking the other links in the candidate network as given. An included new link must be weakly beneficial to both households, with at least one household strictly benefiting under the tie-breaking convention in the main text, while no excluded link is mutually beneficial given the resulting network.

Because the marginal value $\Delta CE_i(ij \mid G)$ depends on the other links contained in G, these bilateral-consent conditions do not guarantee that a pure network satisfying all of them exists for every parameter configuration. Nor do they imply uniqueness when several such networks exist. We therefore do not impose a general pure-strategy network-existence or network-selection result. However, link formation is likely when κ is sufficiently low relative to the informational and insurance benefits of the connection.

Conditional on such a network, period-2 migration choices form a simultaneous-move game. A pure period-2 migration equilibrium is a migration profile in which each household's migration choice satisfies the migration condition derived above, given the migration choices of the other households. We do not require this equilibrium to be unique, and the results below are understood conditional on a continuation migration equilibrium.

Period-1 households evaluate migration and non-migration anticipating the continuation network and period-2 migration behavior. The period-1 equilibrium results below are therefore conditional on such continuation behavior.

Corollary 7 (Multiple Migration Equilibria)**.** *Under suitable parameter restrictions, multiple migration equilibria at $t = 1$ may exist.*

Proof. It is sufficient to provide a parameter configuration under which more than one migration profile is an equilibrium. Consider two identical connected households, i and j, holding the migration decisions of all other households fixed. Because the information generated by household i's own migration and the information obtained from a migrating neighbor are substitutes, the continuation-value gain from $M_{i1} = 1$ is larger when $M_{j1} = 0$ than when $M_{j1} = 1$. For intermediate expected migration returns and migration costs, it can therefore be the case that

$$\phi_{i1}(M_{j1} = 0) > 0, \qquad \phi_{i1}(M_{j1} = 1) < 0.$$

By symmetry, the same restrictions imply

$$\phi_{j1}(M_{i1} = 0) > 0, \qquad \phi_{j1}(M_{i1} = 1) < 0.$$

Hence household i prefers to migrate when j does not migrate and to wait when j migrates, and the same is true for household j. Consequently, both

$$(M_{i1}, M_{j1}) = (1, 0) \qquad \text{and} \qquad (M_{i1}, M_{j1}) = (0, 1)$$

are migration equilibria. Thus multiple period-1 migration equilibria may exist.

Proposition 7 (Empty Network)**.** *Suppose the initial network is empty and households have common priors and symmetric payoffs. Then,*

$$\phi_{i1} = \phi > 0 \quad \Longrightarrow \quad NE = (M, M, \ldots, M),$$

whereas

$$\phi_{i1} = \phi < 0 \quad \Longrightarrow \quad NE = (NM, NM, \ldots, NM).$$

If $\phi = 0$, each household is indifferent between migrating and not migrating, so multiple migration equilibria exist absent a tie-breaking rule.

Remark 1 (Symmetric Networks)**.** *Symmetry of the network and household primitives does not require equilibrium migration profiles to be symmetric. In a symmetric non-empty network, symmetric equilibria with all households migrating or all households not migrating can arise, but asymmetric equilibria can also exist because households' migration decisions are strategically interdependent. In particular, as shown in Corollary 7, when migration decisions are strategic substitutes, identical connected households may prefer to migrate when their neighbors do not migrate and to remain when their neighbors migrate, generating asymmetric migration equilibria.*

B Comparison to Existing Models

This paper relates to several strands of the migration and network formation literature. To preserve focus in the main text, we summarize the key conceptual distinctions here.

Comparison with Morten (2019). Our framework builds on the insight that migration and informal insurance are jointly determined under risk, as in Morten (2019), but differs along three fundamental dimensions.

First, while Morten (2019) studies migration primarily as a temporary insurance mechanism within an exogenously given network, we introduce a two-period dynamic framework in which migration decisions generate information that shapes future behavior. Second, whereas risk sharing is the central force disciplining migration in Morten (2019), our model emphasizes learning about migration returns through social networks. Households update beliefs based on observed urban income and strategically form new links to acquire information. Third, networks in our model are partially endogenous: households form new connections between periods, trading off informational gains against kinship-based redistribution costs.

These features create strategic substitutability in migration choices (Proposition 1), which can generate endogenous migration delay, as well as degree-based partner selection in network formation (Corollary 4), mechanisms absent from Morten (2019).

Relation to Kleemans (2015). Our framework is also related to Kleemans (2015), who studies migration as an investment decision under income risk and liquidity constraints. In Kleemans (2015), migration responses to income shocks are driven by the interaction between fixed migration costs and credit constraints, implying that positive income realizations can facilitate migration by relaxing liquidity constraints.

Our model differs in two key respects. First, we abstract from borrowing constraints and instead emphasize uncertainty over migration returns. Second, information plays no role

in Kleemans (2015), whereas learning about migration returns is central in our framework. Migration outcomes generate signals that affect future beliefs and shape both subsequent migration and endogenous network formation.

As a result, while Kleemans (2015) predicts that positive income shocks can increase migration through liquidity effects, our model predicts that higher income risk can increase migration incentives through self-insurance motives (Proposition 2). These mechanisms are complementary but conceptually distinct.

Relation to the endogenous network formation literature. Our model contributes to the literature on endogenous network formation (Bala and Goyal, 2000, Jackson and Wolinsky, 1996, Hellmann, 2021), but differs in both motivation and informational structure.

In standard models, links are formed to exploit payoff complementarities or externalities in static environments. In contrast, we study endogenous link formation driven by learning under uncertainty. Links are formed after observing migration outcomes and reduce uncertainty about future migration gains.

Relative to empirical work on informal risk-sharing networks (Fafchamps, 2011, Ambrus et al., 2014), our framework endogenizes information acquisition while allowing redistribution costs to discipline network expansion. These mechanisms can generate interior optimal degrees (Proposition 4) and heterogeneous linking patterns across households (Corollary 4).

Relation to Bramoullé and Kranton (2007) and endogenous risk-sharing networks. Our model is also related to Bramoullé and Kranton (2007), who study how network structure affects risk sharing and individual incentives in environments with informal insurance. In their framework, networks shape consumption volatility and welfare through static risk-sharing externalities, and agents' positions in the network determine their exposure to aggregate and idiosyncratic shocks.

Our contribution differs in both mechanism and timing. First, while Bramoullé and Kranton (2007) focuses on risk sharing as the primary role of the network, our framework allows the same links to provide migration information while also generating insurance benefits and redistribution obligation. Households therefore form new links by trading off the option value of information against the financial consequences of the relationship. Second, network structure in our model is endogenous and evolves over time: households form new links after observing migration outcomes and before making subsequent migration decisions. Third, kinship-based redistribution creates an incentive to avoid highly connected partners whose resources are divided across more relationships (Proposition 5 and Corollary 4), a force absent from Bramoullé and Kranton (2007).

Distinction from Song and van der Schaar (2015). The closest theoretical analogue to our framework is Song and van der Schaar (2015), which studies dynamic network formation under incomplete information. While both frameworks feature learning and strategic link formation, the source of uncertainty differs fundamentally.

In Song and van der Schaar (2015), agents are uncertain about partners' types and form links so learn partners' types through repeated interaction. Here benefit of forming a link is uncertain. In contrast, households in our model are ex-ante symmetric, and uncertainty arises from a common but unobserved urban income component and idiosyncratic shocks.

Links in our model are formed not to learn about partners' strategic characteristics, but to obtain signals about uncertain migration returns. The informational value of these signals arises from the option to adapt the subsequent migration decision after observing them, rather than from posterior-variance reduction itself. In the full model, this informational value is combined with the insurance and redistribution consequences of the new relationship (Proposition 5). This distinguishes our mechanism from Song and van der Schaar (2015), where link value depends on learning about the characteristics of the partner itself.